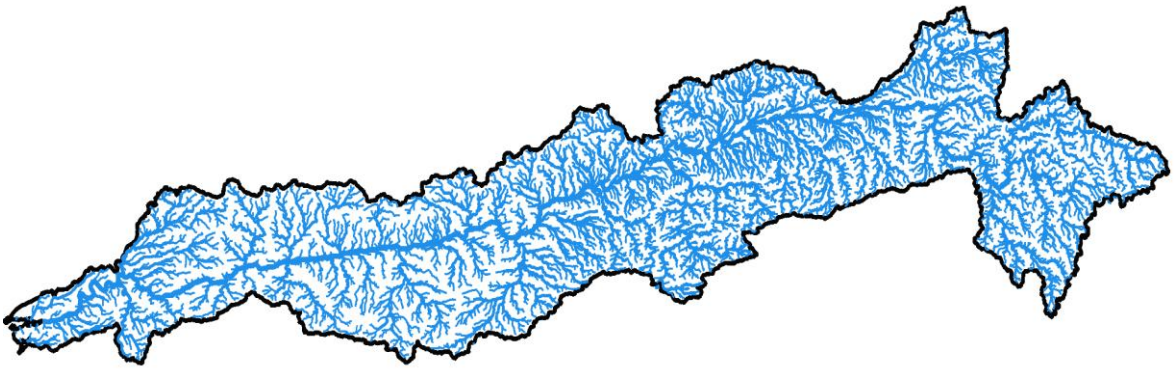
National River Conservation Directorate
Ministry of Jal Shakti, Department of Water Resources,
River Development & Ganga Rejuvenation
Government of India

Pollution Source Mapping for Narmada River Basin



September 2025

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Pollution Source Mapping for Narmada River Basin





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National River Conservation Directorate (NRCD)

The National River Conservation Directorate, functioning under the Department of Water Resources, River Development & Ganga Rejuvenation, and the Ministry of Jal Shakti, provides financial assistance to the State Government for the conservation of rivers under the Centrally Sponsored Schemes of 'National River Conservation Plan (NRCP)'. National River Conservation Plan to the State Governments/ local bodies to set up infrastructure for pollution abatement of rivers in identified polluted river stretches based on proposals received from the State Governments/ local bodies.

www.nrcd.nic.in

Centres for Narmada River Basin Management and Studies (cNarmada)

The Centre for Narmada River Basin Management and Studies (cNarmada) is a Brain Trust dedicated to River Science and River Basin Management. Established in 2024 by IIT Gandhinagar and IIT Indore, under the supervision of cGanga at IIT Kanpur, the centre serves as a knowledge wing of the National River Conservation Directorate (NRCD). cNarmada is committed to restoring and conserving the Narmada River and its resources through the collation of information and knowledge, research and development, planning, monitoring, education, advocacy, and stakeholder engagement.

www.cnarmada.org

Centres for Ganga River Basin Management and Studies (cGanga)

cGanga is a think tank formed under the aegis of NMCG, and one of its stated objectives is to make India a world leader in river and water science. The Centre is headquartered at IIT Kanpur and has representation from most leading science and technological institutes of the country. cGanga's mandate is to serve as a think-tank in the implementation and dynamic evolution of the Ganga River Basin Management Plan (GRBMP) prepared by the Consortium of 7 IITs. In addition to this, it is also responsible for introducing new technologies, innovations, and solutions into India.

www.cganga.org

Acknowledgment

This report is a comprehensive outcome of the project jointly executed by IIT Gandhinagar (Lead Institute) and IIT Indore (Fellow Institute) under the supervision of cGanga at IIT Kanpur. It was submitted to the National River Conservation Directorate (NRCD) in 2024. We gratefully acknowledge the individuals who provided information and photographs for this report.

Team Members

PREFACE

The Narmada River Basin, often referred to as the lifeline of central India, sustains millions of people, ecosystems, and economic activities. With rapid urbanization, industrial growth, and increasing demands on natural resources, the basin faces significant challenges related to pollution, wastewater management, and waste generation.

This report has been prepared to provide a comprehensive assessment of the current status of water quality and pollution sources across the upper and middle stretches of the basin. It brings together data from monitoring stations, evaluates industrial and domestic wastewater loads, and reviews the performance of sewage treatment plants (STPs) and other treatment facilities. Special focus has been given to various categories of waste — including solid, hazardous, biomedical, plastic, construction & demolition (C&D), and e-waste — to understand their generation patterns .

The objective of this document is to support informed decision-making for policymakers, administrators, researchers, and stakeholders engaged in river basin management. It aims to strengthen ongoing initiatives such as the Swachh Bharat Mission, Smart Cities Mission, and Nirmal Dhara, while contributing to the broader goals of sustainable development and ecological preservation.

We extend our sincere gratitude to the government agencies, research institutions, project staff, and individuals whose contributions of data and insights made this assessment possible.

It is our hope that this report inspires informed dialogue and coordinated action, contributing to the restoration and long-term sustainability of the Narmada River Basin for both ecological health and socio-economic development.

Centre for Narmada River Basin
Management and Studies (cNarmada)
IIT Gandhinagar, IIT Indore

Table of Contents

1 1

2 2

2.1 2

2.1.1 2

2.1.2 5

3 9

3.1 9

3.1.1 9

3.1.2 10

3.2 12

3.2.1 12

3.2.2 13

3.3 14

3.3.1 14

3.3.2 16

4 17

4.1 17

4.1.1 17

4.1.1.1 **Error! Bookmark not defined.**

4.1.1.2 18

4.1.1.3 19

4.1.1.4 20

4.1.2 22

4.1.2.1 22

4.1.2.2 23

4.1.2.3 24

4.1.2.4 26

5 28

5.1 28

5.1.1 28

5.2 29

6 31

6.1 31

6.2 32

7 34

7.1 34
7.2 35
8 37
8.1 37
8.2 38
9 40
9.1 40
9.2 41
10 43
10.1 43
10.2 44
11 46
References 46

List of Figures

Figure 1 Spatial distribution of water quality index parameters at monitoring stations along the Narmada Upper Basin 2
Figure 2. Spatial distribution of water quality index parameters at monitoring stations along the Narmada Middle Basin 5
Figure 3 Upper Narmada River Basin District-wise Distribution of Red, Orange, and Green Category Industries 9
Figure 4 Middle Narmada River Basin District-wise Distribution of Red, Orange, and Green Category Industries 11
Figure 5 District-wise Domestic Sewage Generation (MLD) in Narmada Upper Basin 12
Figure 6 District-wise Domestic Sewage Generation (MLD) in Narmada Middle Basin 13
Figure 7 District-wise Industrial Sewage Generation (MLD) in Narmada Upper Basin 15
Figure 8 District-wise Industrial Sewage Generation (MLD) in Narmada Middle Basin 16
Figure 9 Number of Operational STPs per district in Narmada Upper Basin 17
Figure 10 Number of Under Construction STPs per district in Narmada Upper Basin 19
Figure 11 District-wise cumulative capacity (MLD) of Operational STPs in Narmada Upper Basin 20
Figure 12 District-wise cumulative capacity (MLD) of Under Construction STPs in Narmada Upper Basin 21
Figure 13 District-wise Operational STPs in Narmada Middle Basin 22
Figure 14 District-wise Under-Construction STPs in Narmada Middle Basin 24
Figure 15 District-wise cumulative capacity (MLD) of Operational STPs in Narmada Middle Basin 25
Figure 16 District-wise cumulative capacity (MLD) of under-construction STPs in Narmada Middle Basin 27
Figure 17 District-wise Solid Waste Generation (TPD) in the Narmada Upper Basin 28
Figure 18District-wise Solid Waste Generation (TPD) in the Narmada Middle Basin 29
Figure 19 District-wise Hazardous Waste Generation (TPD) in the Narmada Upper Basin 31
Figure 20District-wise Hazardous Waste Generation (TPD) in the Narmada Middle Basin 32
Figure 21 District-wise Biomedical Waste Generation (TPD) in the Narmada 34
Figure 22 District-wise Biomedical Waste Generation (TPD) in the Narmada Middle Basin 36
Figure 23 District-wise Plastic Waste Generation (TPD) in the Narmada Upper Basin 37
Figure 24 District-wise Plastic Waste Generation (TPD) in the Narmada Middle Basin 38

Figure 25 District-wise Construction and Demolition Waste Generation (TPD) in the Narmada Upper Basin, highlighting major and minor hotspots. 40
Figure 26 District-wise Construction and Demolition Waste Generation (TPD) in the Narmada Middle Basin, highlighting major and minor hotspots. 41
Figure 27 District-wise Electronic Waste Generation in the Narmada Upper Basin, highlighting hotspots and data gaps 43
Figure 28 District-wise Electronic Waste Generation in the Narmada Middle Basin, highlighting hotspots and data gaps. 45

List of Tables

Table 1 Water Quality Parameters Standard Limits 3
Table 2 List of MPPCB Monitoring Stations in Upper Basin 3
Table 3 List of MPPCB Monitoring Stations in Middle Basin 6

Abbreviations and Acronyms

BOD – Biochemical Oxygen Demand

C&D Waste – Construction and Demolition Waste

COD – Chemical Oxygen Demand

CPCB – Central Pollution Control Board

DO – Dissolved Oxygen

MLD – Million Litres per Day

MPPCB – Madhya Pradesh Pollution Control Board

MSW – Municipal Solid Waste

NGT – National Green Tribunal

NRCD – National River Conservation Directorate

NRCP – National River Conservation Plan

NMCG – National Mission for Clean Ganga

pH – Potential of Hydrogen

STP – Sewage Treatment Plant

TDS – Total Dissolved Solids

F.C. – Fecal Coliforms

1 Introduction

The Narmada River is one of India's most vital west-flowing rivers, supporting agriculture, drinking water supply, industry, and ecosystems across Madhya Pradesh, Maharashtra, and Gujarat along with smaller parts of Chhattisgarh and Maharashtra. The Narmada River travels through Madhya Pradesh (1,077 km), Maharashtra (74 km), including along the border between Madhya Pradesh and Maharashtra (39 km), then the border between Maharashtra and Gujarat (74 km), and finally Gujarat (161 km).

However, rapid urbanization, industrialization, and population growth have increasingly exerted pressure on the river's water quality and ecological health. The discharge of untreated domestic sewage, **industrial effluents**, and the mismanagement of **solid waste, hazardous waste, biomedical waste, plastics, construction & demolition (C&D) waste, and e-waste** have emerged as major contributors to pollution in the basin. These wastes, when disposed of improperly, introduce pollutants such as organic matter, nutrients, heavy metals, persistent organic pollutants, plastics, and other non-biodegradable materials into the river system.

In order to assess the current status of pollution in the Narmada River Basin, secondary data from multiple government agencies has been compiled and analyzed. These include the Madhya Pradesh Pollution Control Board (MPPCB), Center Pollution Control Bord (CPCB), Ministry of MSME, records from the National Green Tribunal (NGT) Government of Madhya Pradesh and other relevant agencies.

By integrating secondary datasets from regulatory agencies this report aims to provide insights into the presence and concentration of key pollutants, trends in water quality parameters. The study evaluates key pollutants such as domestic and industrial wastewater, solid and hazardous waste, biomedical and plastic waste, and e-waste while also mapping of existing and under-construction sewage treatment facilities is also there.

This basin-wide data provides crucial insights into the spatial distribution of pollution, helping identify priority areas where interventions are urgently needed to protect the river's ecological and social functions.

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2 Water Quality Status

2.1 Monitoring Stations, Locations, and Water Quality

2.1.1 Upper Basin

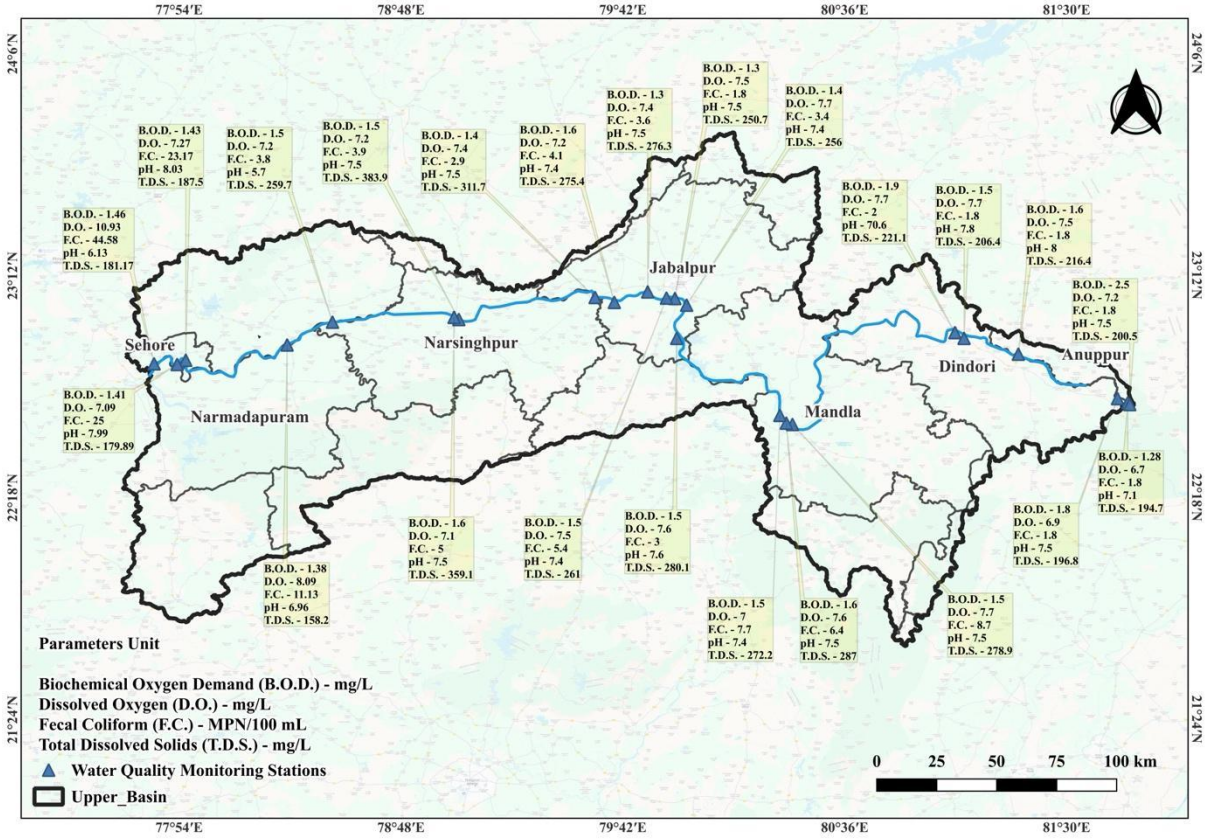


Figure 1 Spatial distribution of water quality index parameters at monitoring stations along the Narmada Upper Basin

(Source: Locations are digitized using Google Maps after taking reference of locations from MPPCB Narmada Water Quality Report 2024-25 and Data is taken from Narmada River monitoring Report 2025, MPPCB)

The Map shown in Fig. 1 provides a spatial overview of water quality in the Narmada Upper Basin, using data from Madhya Pradesh Pollution Control Board (MPPCB) water quality monitoring stations distributed throughout the basin. The map summarizes key water quality indicators at each station: Biochemical Oxygen Demand (BOD), Dissolved Oxygen (DO), Fecal Coliform (FC), pH, and Total Dissolved Solids (TDS). BOD values generally range from 1.3 to 2.5 mg/L, DO fluctuates from around 6.4 to 8.9 mg/L, and TDS varies widely from about 179 to 383 mg/L. Higher BOD and FC values are observed closer to urban centers like Jabalpur and Narmadapuram, suggesting increased organic and fecal contamination in these segments, likely due to inputs from municipal sewage and anthropogenic activities.

Upstream Stations or forested regions, such as near Mandla and Anuppur, typically exhibit lower BOD and higher DO, reflecting less impacted stretches.

Table 1 provides the standard limits of the parameters suggested by different agencies as a guideline for the users to ensure the water quality for direct and indirect uses. The MPPCB water quality monitoring stations, district-wise, in the upper basin are listed in Table 2.

Table 1 Water Quality Parameters Standard Limits

Parameter	Requirement (Acceptable Limit)	Permissible Limit in Absence of Alternate Source	Unit	Agency
TDS	500	2000	mg/L	BIS IS 10500: 2012
pH	6.5-8.5	-		BIS IS 10500: 2013
DO	6 or more	-	mg/L	CPCB
BOD	3 or less	-	mg/L	CPCB
F. Coliform	0	-	MPN/100	WHO

Table 2 List of MPPCB Monitoring Stations in Upper Basin

S. No.	District	City/ Town	Name of Monitoring Station	B. O. D	Dissolved Oxygen	Fecal Coliform	pH	TDS
1	Anuppur	Amarkantak	Narmada River at Amarkantak origin point, Dist. Anuppur	1.28	6.7	1.8	7.1	194.7
2	Anuppur	Amarkantak	Narmada River at Puskar Dam, Amarkantak	2.5	7.2	1.8	7.5	200.5
3	Anuppur	Amarkantak	Narmada River at Kapildhara	1.8	6.9	1.8	7.5	196.8
4	Dindori	Dindori	Narmada River at Chandanghat near Road Bridge, Dist. Dindori	1.6	7.5	1.8	8	216.4
5	Dindori	Dindori	Narmada River at Dindori Up Stream	1.5	7.7	1.8	7.8	206.4
6	Dindori	Dindori	Narmada River at Dindori Down Stream	1.9	7.7	2	70.6	221.1
7	Mandla	Mandla	Narmada River near Shamshanghat, Mandla	1.5	7.7	8.7	7.5	278.9

8	Mandla	Mandla	Narmada River near Road Bridge, Mandla	1.6	7.6	6.4	7.5	287
9	Mandla	Mandla	Narmada River at Bhairav Temple, Shhastradhar, Mandla	1.5	7	7.7	7.4	272.2
10	Jabalpur	Jabalpur	Narmada River near Road Bridge (Down Stream of Bargi Dam), Jabalpur	1.5	7.6	3	7.6	280.1
11	Jabalpur	Jabalpur	Narmada River at Jamtara, near Railway Bridge	1.4	7.7	3.4	7.4	256
12	Jabalpur	Lalpur	Narmada River near Water Supply Intake Point, Lalpur	1.3	7.5	1.8	7.5	250.7
13	Jabalpur	Jabalpur	Narmada River at Tilwaraghat	1.5	7.5	5.4	7.4	261
14	Jabalpur	Jabalpur	Narmada River at Panchwatighat, Bheraghat	1.3	7.4	3.6	7.5	276.3
15	Jabalpur	Jabalpur	Narmada River at Sarswastighat after mixing Bawanganga River	1.6	7.2	4.1	7.4	275.4
16	Jabalpur	Jabalpur	Narmada near Road Bridge at Jhansinghat, Jabalpur	1.4	7.4	2.9	7.5	311.7
17	Narsinghpur	Narsinghpur	Narmada N.H.44 near Road Bridge at Barman, Jabalpur	1.5	7.2	3.9	7.5	383.9
18	Narsinghpur	Narsinghpur	Narmada Barmanghat, 100 mts D/s Mainghat	1.6	7.1	5	7.5	359.1
19	Narsinghpur	Narsinghpur	Narmada S.H.44 near Road Bridge, Jhilkoli, Narsinghpur	1.5	7.2	3.8	5.7	259.7
20	Narmadapuram	Sandia	Narmada River near Road Bridge, Sandia, Narmadapuram	1.38	8.09	11.13	6.96	158.2
21	Sehore	Jait	Narmada River at Up Stream of Village Jait, Sehore	1.43	7.27	23.17	8.03	187.5
22	Sehore	Jait	Narmada River at Down Stream of Village Jait, Sehore	1.41	7.09	25	7.99	179.89

23	Sehore	Sehore	Narmada River at Shahaganj Guest House, Sehore	1.46	10.93	44.58	6.13	181.17

(Source: MPPCB Narmada Water Quality Report 2024-25)

2.1.2 Middle Basin

The Map shown in Fig. 2 offers an integrated snapshot of water quality in the Narmada Middle Basin using various monitoring stations along the river. Each station reports Biochemical Oxygen Demand (BOD), Dissolved Oxygen (DO), Fecal Coliform (FC), pH, and Total Dissolved Solids (TDS). Most BOD values remain below 2 mg/L, signaling moderate organic pollution. DO levels generally range from approximately 7 to 8 mg/L, which is favorable for aquatic life. FC bacteria counts and TDS indicate localized contamination and mineral load, revealing spatial variability along the river. Stations near urban centers like Sehore and Narmadapuram tend to have higher BOD and FC values, pointing to significant organic and microbial pollution inputs from population and development pressures. In Khandwa , Dhar BOD stays low, and DO remains relatively high, suggesting healthier river stretches with fewer anthropogenic disturbances. The data-rich, spatial view allows officials to pinpoint critical pollution hotspots and tailor water quality interventions. Prioritizing sanitation projects and continuous monitoring in high BOD/FC areas will improve river health for communities and ecosystems across the Narmada Middle Basin. The MPPCB water quality monitoring stations district wise in middle basin are listed in Table 3.

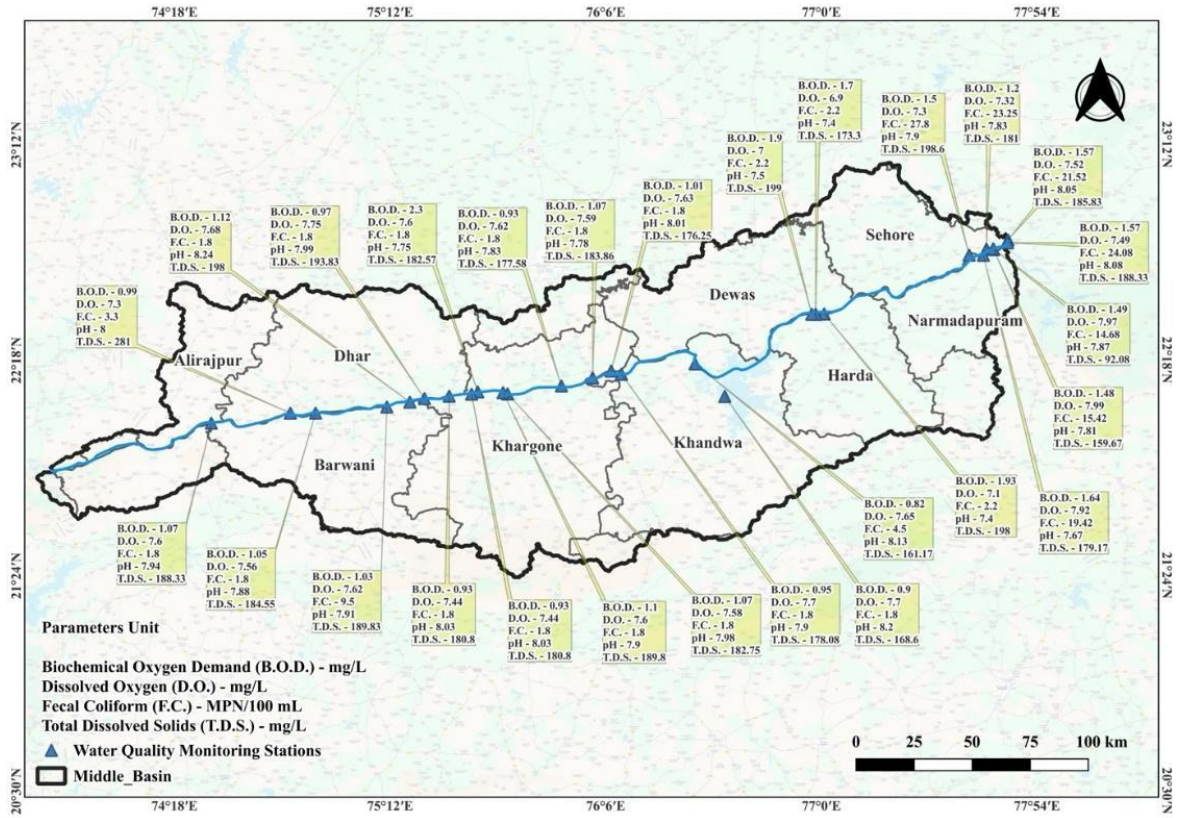


Figure 2. Spatial distribution of water quality index parameters at monitoring stations along the Narmada Middle Basin

(Source: Locations are digitized using Google Maps after taking reference of locations from MPPCB Narmada Water Quality Report 2024-25 and Data is taken from Narmada River monitoring Report 2025, MPPCB)

Table 3 List of MPPCB Monitoring Stations in Middle Basin

S. No	District	City/ Town	Name of Monitoring Station	B.O.D	Dissolved Oxygen	Fecal Coliform	pH	TDS
1	Sehore	Sehore	Narmada Up Stream before Confluence of River Tawa near Bandrabhan, Sehore	1.57	7.52	21.52	8.1	185.8
2	Sehore	Ramnagar	Narmada Down Stream after Confluence of River Tawa, Ramnagar	1.57	7.49	24.08	8.1	188.3
3	Sehore	Budhni	Narmada Budhnighat, Budhni	1.2	7.32	23.25	7.8	181
4	Narmadapuram	Narmadapuram	Narmada Korighat, Hoshangabad	1.48	7.99	15.42	7.8	159.7

5	Narmadapuram	Narmadapuram	Narmada Sethanighat, Hoshangabad	1.49	7.97	14.68	7.9	92.08
6	Narmadapuram	Narmadapuram	Narmada River 100 m Down Stream after SPM Nalla	1.64	7.92	19.42	7.7	179.2
7	Sehore	Budhni	Narmada at Down Stream of Textile Unit, Village Holipura, Budhni	1.5	7.3	27.8	7.9	198.6
8	Dewas	Nemawar	Narmada at before Confluence of River Jamner, Village Nemawar	1.93	7.1	2.2	7.4	198
9	Dewas	Nemawar	Narmada at Water Supply Intake Point, Nemawar	1.7	6.9	2.2	7.4	173.3
10	Dewas	Nemawar	Narmada 500 m Down Stream near Jain Mandir, Village Nemawar	1.9	7	2.2	7.5	199
11	Khandwa	Hanuwantia	Narmada at Hanuwantia, Dist. Khandwa	0.9	7.7	1.8	8.2	168.6
12	Khandwa	Punasa	Narmada River at Punasa Dam	0.82	7.65	4.5	8.1	161.2
13	Khandwa	Omkareshwar	Narmada at Up Stream, Omkareshwar Dam	0.95	7.7	1.8	7.9	178.1
14	Khandwa	Omkareshwar	Narmada at Omkareshwar Down Stream	1.01	7.63	1.8	8	176.3
15	Khargone	Barwaha	Narmada, Barwaha near Mortakka Bridge	1.07	7.59	1.8	7.8	183.9
16	Khargone	Dhareshwar	Narmada River at Dhareshwar	0.93	7.62	1.8	7.8	177.6
17	Khargone	Jallod	Narmada near Water Supply Intake, Jallod	1.07	7.58	1.8	8	182.8

18	Khargone	Mandleshwar	Narmada River at Mandleshwar Down Stream	1.1	7.6	1.8	7.9	189.8
19	Khargone	Maheshwar	Narmada River at Maheshwar Down Stream	2.3	7.6	1.8	7.8	182.6
20	Khargone	Maheshwar	Narmada River at Shahastradhara (Jalkoti)	0.93	7.44	1.8	8	180.8
21	Dhar	Dhar	Narmada at Khalghat	0.93	7.44	1.8	8	180.8
22	Dhar	Dharampuri	Narmada at Dharampuri	0.97	7.75	1.8	8	193.8
23	Dhar	Dharampuri	Narmada River at Dharampuri Down Stream	1.12	7.68	1.8	8.2	198
24	Barwani	Barwani	Narmada River at Semalda Up Stream of Barwani	1.03	7.62	9.5	7.9	189.8
25	Barwani	Barwani	Narmada River at Rajghat, Barwani	1.05	7.56	1.8	7.9	184.6
26	Dhar	Nisarpur	Narmada River at Koteshwar, Nisarpur, Dhar	0.99	7.3	3.3	8	2811
27	Aliraj pur	Kakrana	Narmada River at Kakrana	1.07	7.6	1.8	7.9	188.3

(Source: MPPCB Narmada Water Quality Report 2024-25)

3 Pollution Sources

3.1 Industrial Clusters

3.1.1 Upper Basin

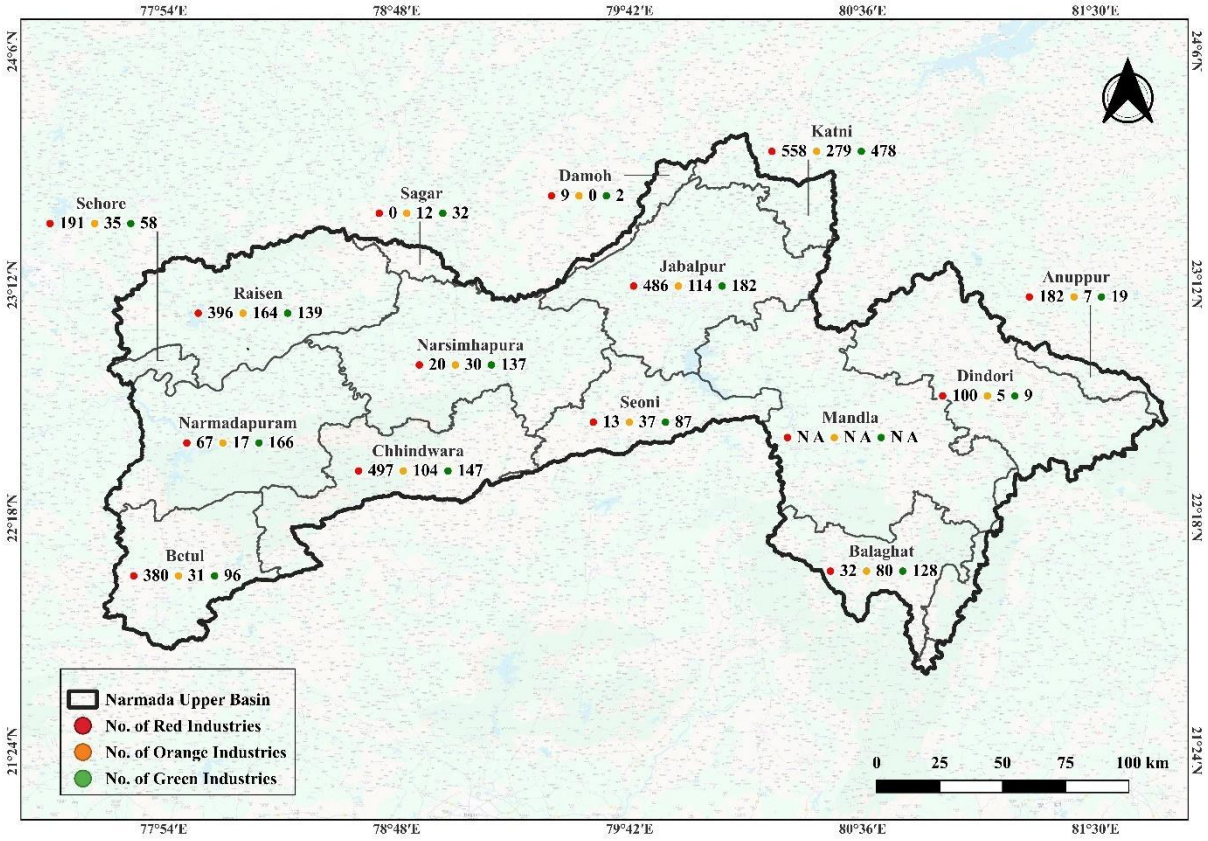


Figure 3 Upper Narmada River Basin District-wise Distribution of Red, Orange, and Green Category Industries

(Source: Madhya Pradesh Pollution Control Board (MPPCB). (2025). District Environmental Plans (DEPs))

The industrial ecosystem of the Upper Narmada River basin, shown in Fig. 3, demonstrates substantial district-level heterogeneity in the number and type of industrial units, as classified by the Central Pollution Control Board in Red, Orange, and Green categories. Katni emerges as the most industrialized district in the basin, registering 558 red, 279 orange, and 478 green industries, reflecting a broad and intense industrial base with likely significant environmental pressures. This is closely followed by Jabalpur and Chhindwara, with Jabalpur housing 486 red, 114 orange, and 182 green units, while Chhindwara has 497 red, 104 orange, and 147 green units. Raisen and Betul are notable for their high figures as well, evidencing considerable manufacturing and processing activities distributed across their territories.

In contrast, districts such as Sehore, Anuppur, Balaghat, and Narsimhapur, while still actively industrial, have lower absolute numbers, with green-category industries often outnumbering orange and red, suggesting either a prevalence of less polluting enterprises or a strategic shift toward environmentally friendly industrial practices. Districts such as Sagar and Damoh present a minimal red industry and a clear dominance of less hazardous (green and orange) sectors, indicating localized efforts or structural economic factors favoring lower-impact industry. Mandla stands out, showing no available data or negligible industrial activity across all three categories, potentially due to ecological prioritization, stricter land-use regulations, or reporting gaps.

This spatial variation underscores the dynamic nature of industrial development within the Upper Narmada Basin. Some districts serve as primary industrial corridors, potentially contributing heavily to both economic growth and cumulative pollution loads, while others remain less industrialized, either because of policy, infrastructure, or geography. These patterns are critical for regional planning, permitting targeted regulatory measures and resource allocation to manage and mitigate industrial pollution, especially in high-concentration red category zones. The basin-wide industrial profile highlights the necessity for context-sensitive interventions to balance economic advancement with sustainable water and environmental stewardship in the Narmada region.

3.1.2 Middle Basin

The Narmada Middle Basin exhibits a diverse industrial profile, with pronounced variation across its constituent districts in terms of both the quantity and environmental categorization of industrial units The Map shown in Fig. 4. Indore stands out as the foremost industrial hub within the basin, hosting 276 red, 473 orange, and a remarkable 941 green category industries. This reflects both significant economic activity and a progressive shift towards less polluting, green industries. Dhar also demonstrates substantial industrial density, with 351 red, 270 orange, and 734 green industries, highlighting its role in the region's manufacturing and processing sectors.

Several districts, including Raisen, Betul, Chhota Udepur, and Dewas, display considerable numbers of red and orange industries, suggesting a strong presence of high and moderate pollution potential activities, balanced by notable green category establishments. Alirajpur and Jhabua, by contrast, have modest totals, with a predominance of green industries relative to

their red and orange counterparts, indicating either an orientation towards environmentally conscious enterprises or limited large-scale industry.

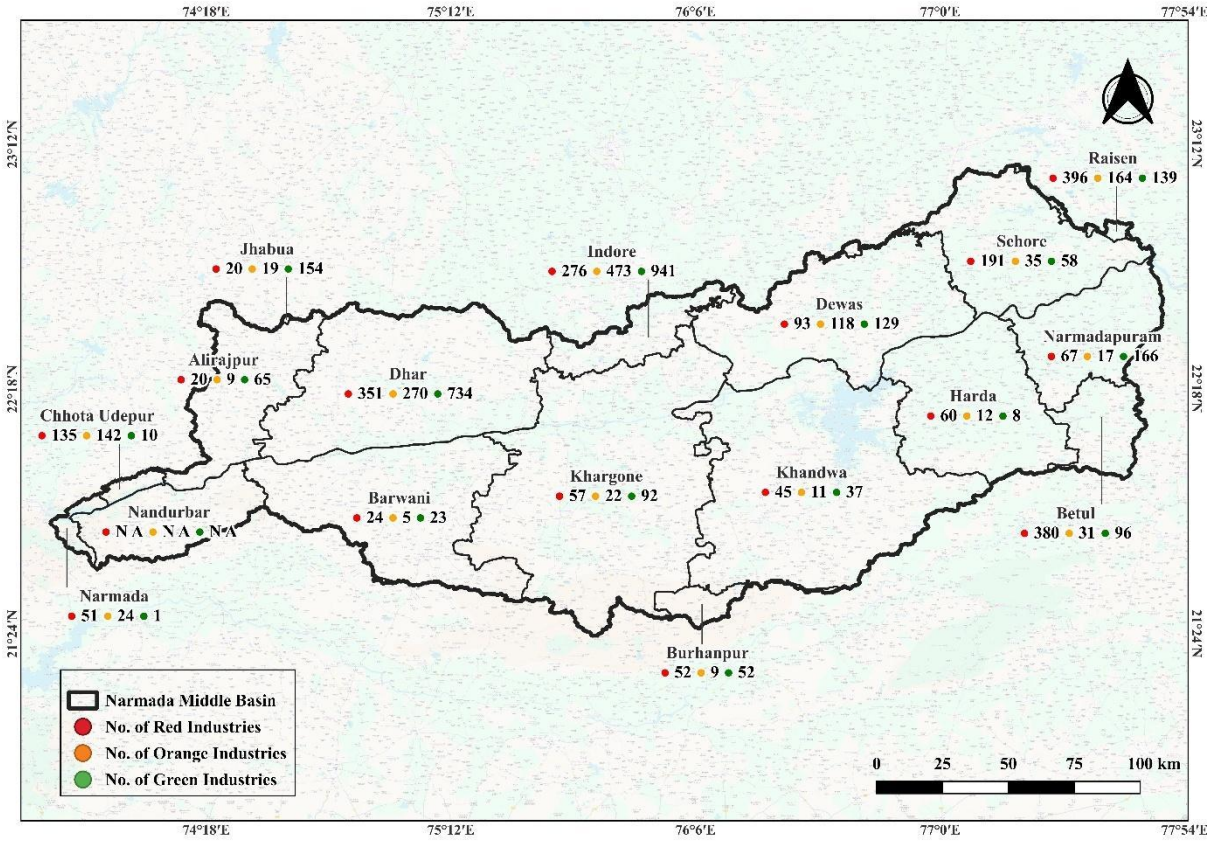


Figure 4 Middle Narmada River Basin District-wise Distribution of Red, Orange, and Green Category Industries

(Source: Madhya Pradesh Pollution Control Board (MPPCB). (2025). District Environmental Plans (DEPs))

Some districts, such as Nandurbar, exhibit negligible or unavailable data for all three categories, potentially pointing to gaps in reporting, the absence of significant industrial operations, or preferential policies favoring low-impact development. Across Harda, Khargone, Burhanpur, and other districts, the trend of green industries outnumbering red and orange ones persists, signifying efforts towards sustainable industrialization.

This spatial and categorical differentiation underscores the varied environmental and economic landscape of the Narmada Middle Basin. Districts with high concentrations of red-category units may be subject to greater regulatory focus and pollution mitigation requirements, while areas favoring green industries illustrate a move towards cleaner, sustainable development. The observed distribution of industry types is instrumental for planning targeted interventions, resource prioritization, and fostering environmentally responsible growth across the basin.

3.2 Domestic Wastewater Load

3.2.1 Upper Basin

The Map shown in Fig. 5 illustrates the spatial distribution of domestic sewage generation across districts within the Narmada Upper Basin region of Madhya Pradesh, India, highlighting significant disparities in sewage loads among various administrative zones. The map uses a color gradient to depict the magnitude of domestic sewage generation in Million Liters Per Day (MLD) for each district, with darker shades representing higher sewage loads and lighter shades indicating comparatively lower contributions.

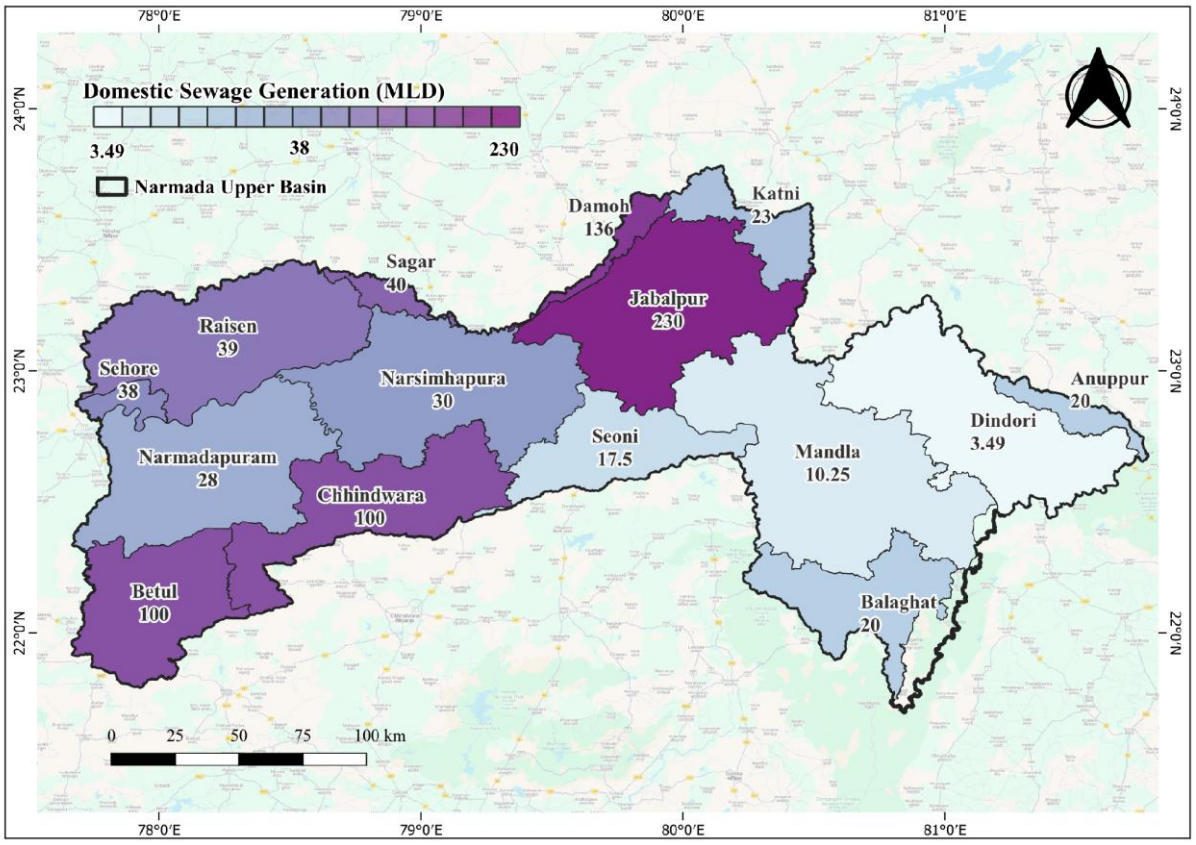


Figure 5 District-wise Domestic Sewage Generation (MLD) in Narmada Upper Basin

(Source: Madhya Pradesh Pollution Control Board (MPPCB). (2025). District Environmental Plans (DEPs))

Jabalpur emerges as the district with the highest generation of domestic sewage at 230 MLD, signifying its urban dominance and relatively large population base. Other key urban centers, such as Betul, Chhindwara, and Damoh, also show elevated sewage outputs of 100 MLD and 136 MLD, respectively, indicating their roles as regional urban and commercial hubs.

Districts such as Seoni (17.5 MLD), Mandla (10.25 MLD), Dindori (3.49 MLD), and Anuppur (20 MLD) are characterized by lighter shades, signifying their relatively low domestic sewage generation rates. These patterns generally reflect smaller population concentrations, lower urbanization levels, and possibly different water usage behaviors. Central districts like Raisen (39 MLD), Sagar (40 MLD), and Narsimhapura (30 MLD) register moderate values, indicative of balanced urban-rural characteristics. A clear correlation is observed between urbanization, population density, and sewage load; districts with major urban centers such as Jabalpur, Betul, and Chhindwara considerably outpace their rural or semi-urban counterparts. This highlights the vital need for robust sewage management and treatment infrastructure in these high-load zones to prevent environmental and public health risks. Conversely, districts with lower MLD values may prioritize decentralized wastewater management and pollution prevention strategies suitable for smaller population clusters.

3.2.2 Middle Basin

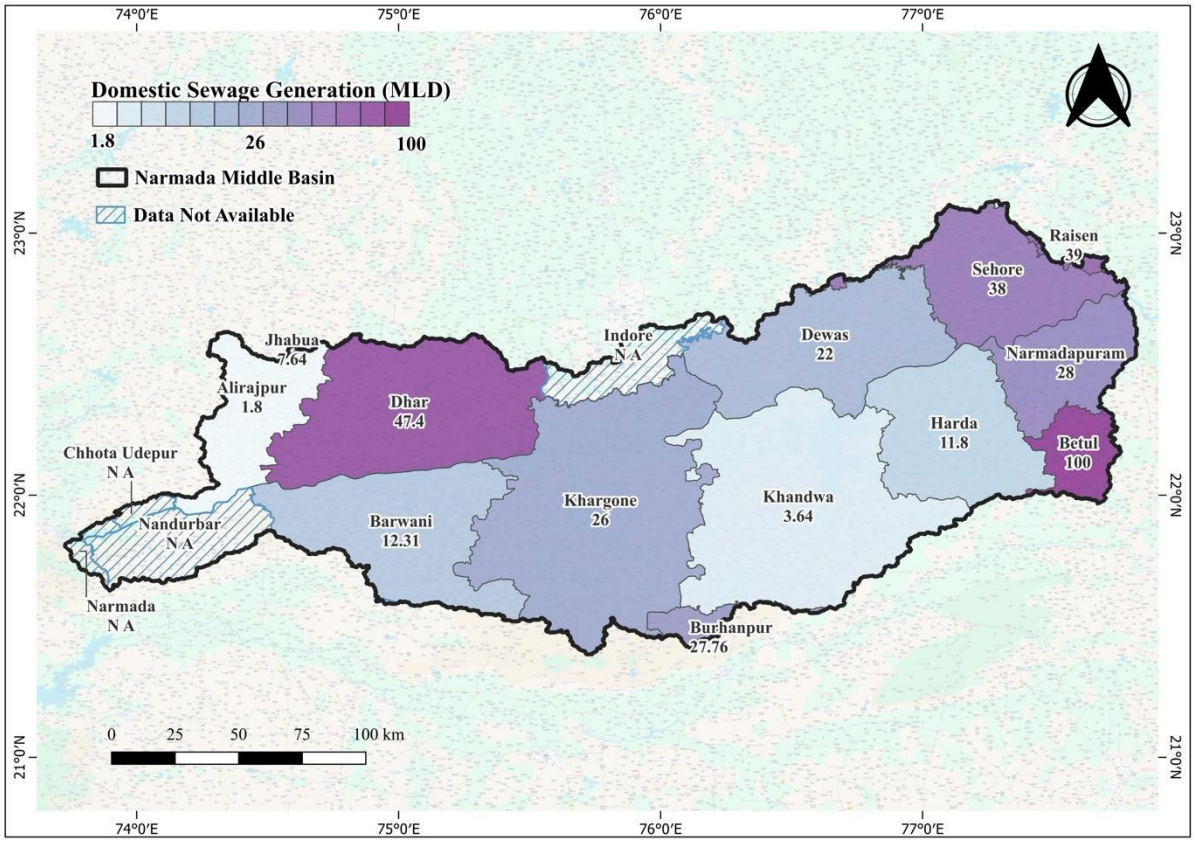


Figure 6 District-wise Domestic Sewage Generation (MLD) in Narmada Middle Basin

(Source: Madhya Pradesh Pollution Control Board (MPPCB). (2025). District Environmental Plans (DEPs))

The Map shown in Fig. 6 illustrates the spatial distribution of domestic sewage generation, measured in Million Liters Per Day (MLD), across the districts falling within the Middle Narmada Basin. The basin boundary is clearly demarcated, and regions with unavailable data are represented using hatched shading. The values reveal significant inter-district variability, reflecting differences in population density, urbanization, and developmental patterns. Among the districts, Betul (100 MLD) emerges as the highest contributor of sewage, followed by Dhar (47.4 MLD), both of which are highlighted in darker purple to denote their considerable wastewater burden. Moderate contributors include Sehore (38 MLD), Raisen (39 MLD), Narmadapuram (28 MLD), Burhanpur (27.76 MLD), and Khargone (26 MLD), reflecting semi-urban settlements and growing towns with intermediate levels of wastewater generation. Districts with relatively low sewage output include Khandwa (3.64 MLD), Alirajpur (1.8 MLD), and Harda (11.8 MLD), typically representing smaller urban centers or predominantly rural populations. Certain districts, notably Indore, Nandurbar, and Narmada, are marked as "Data Not Available," which is a critical limitation given Indore's prominence as a major urban hub within the basin. Spatially, the western districts such as Alirajpur, Jhabua, and Barwani show relatively low sewage generation, consistent with their lower population density, whereas the central and eastern parts, including Dhar, Betul, Raisen, Sehore, and Narmadapuram, emerge as major sewage contributors, reflecting higher urban influence. The southern belt, comprising Burhanpur, Khandwa, and Harda, shows mixed patterns, with Burhanpur producing comparatively higher sewage than its neighboring districts. These observations highlight the urgent need for robust wastewater treatment infrastructure in high-generation districts like Betul and Dhar, while moderate-load areas could benefit from decentralized treatment solutions appropriate for semi-urban contexts. The absence of reliable data for Indore underscores a major research and planning gap, and comprehensive, continuous monitoring remains essential for developing effective sewage management strategies to safeguard the water quality of the Narmada River and its tributaries.

3.3 Industrial Wastewater Load

3.3.1 Upper Basin

The Map shown in Fig. 7 displays the district-wise distribution of industrial sewage generation within the Narmada Upper Basin of Madhya Pradesh, India, emphasizing the substantial variations in industrial wastewater output across the region. The map employs a color spectrum ranging from light yellow to deep red to represent industrial sewage generation in Million Liters

Per Day (MLD). Darker shades indicate districts with higher industrial sewage loads, while lighter shades reflect lower outputs. Narsimhapura stands out with an exceptionally high value of 1006 MLD, signifying it as a major industrial center in the basin. Chhindwara (46.33 MLD) and Betul (51.99 MLD) also show significantly elevated levels, corresponding to the presence of numerous industrial facilities and activities.Contrasting the industrial heavyweights, districts such as Mandla (0.9 MLD), Katni (0.14 MLD), and Dindori (0 MLD) demonstrate minimal or negligible industrial sewage generation, mirroring their limited industrialization and predominance of agriculture or rural livelihoods. Other areas like Sagar (15.3 MLD), Sehore (22 MLD), and Narmadapuram (25 MLD) register moderate output, indicative of localized industrial clusters or small-scale manufacturing units. The spatial concentration of industrial sewage generation follows industrialization patterns within the basin. Districts with high industrial activity, especially Narsimhapura, Betul, and Chhindwara, require robust strategies for industrial effluent treatment and regulatory oversight to mitigate environmental pollution risks. The absence or low levels of industrial sewage in certain districts like Damoh, Mandla, and Dindori hint at their potential suitability for conservation-driven initiatives and lower pollution management needs.

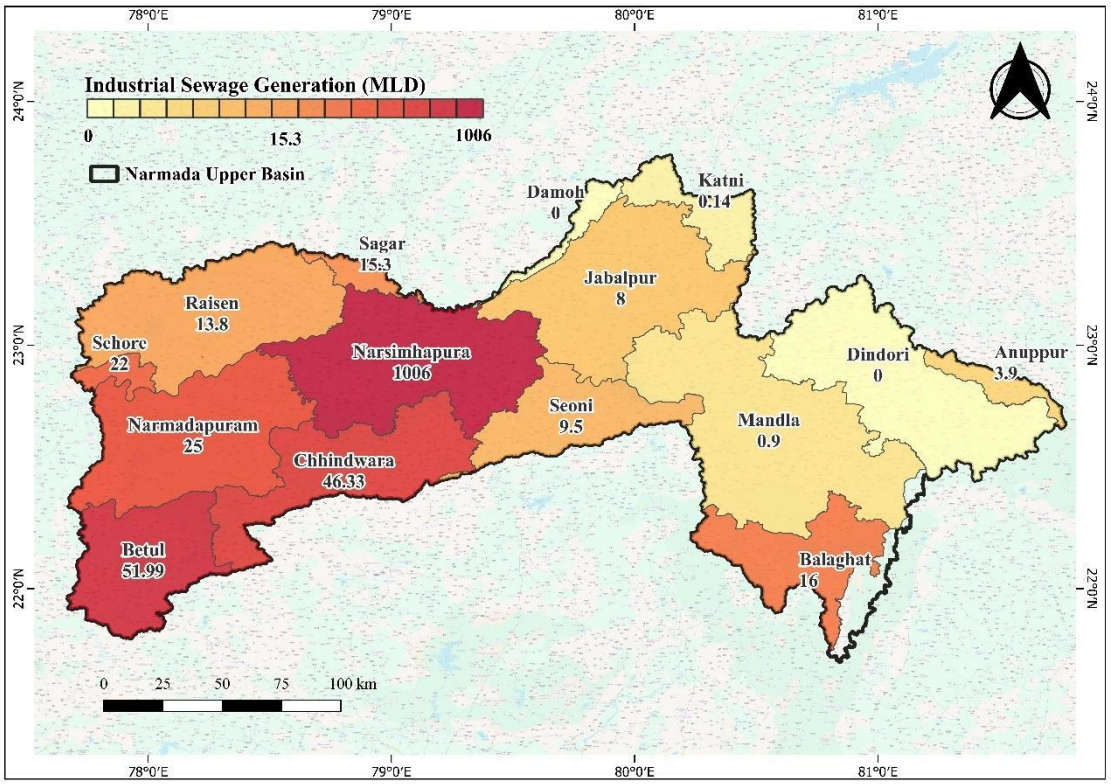


Figure 7 District-wise Industrial Sewage Generation (MLD) in Narmada Upper Basin

(Source: Madhya Pradesh Pollution Control Board (MPPCB). (2025). District Environmental Plans (DEPs))

3.3.2 Middle Basin

The Map shown in Fig. 8 illustrates industrial sewage generation (in MLD, Million Liters per Day) across various districts of the Narmada Middle Basin, categorizing regions according to the volume of sewage generated. Highest emissions are observed in Betul (51.99 MLD), followed by Khandwa (39.96 MLD) and Narmadapuram (25 MLD), while some regions like Harda (0.11 MLD) and Jhabua (0.72 MLD) show minimal contributions.

Betul leads with the highest recorded value of 51.99 MLD, marking it as a critical point source for industrial effluent within the basin. Districts such as Khandwa (39.96 MLD), Narmadapuram (25 MLD), Khargone and Shajapur (22 MLD each), and Dhar (18.6 MLD) also demonstrate significant industrial sewage contributions. Harda (0.11 MLD), Jhabua (0.72 MLD), and Burhanpur (1.07 MLD) represent districts with notably lower industrial sewage outputs, indicating limited industrial activity or effective waste management. Industrial sewage data is not available for certain districts, including Indore, Alirajpur, Nandurbar, and Chhota Udepur, which are indicated with shaded regions on the map.

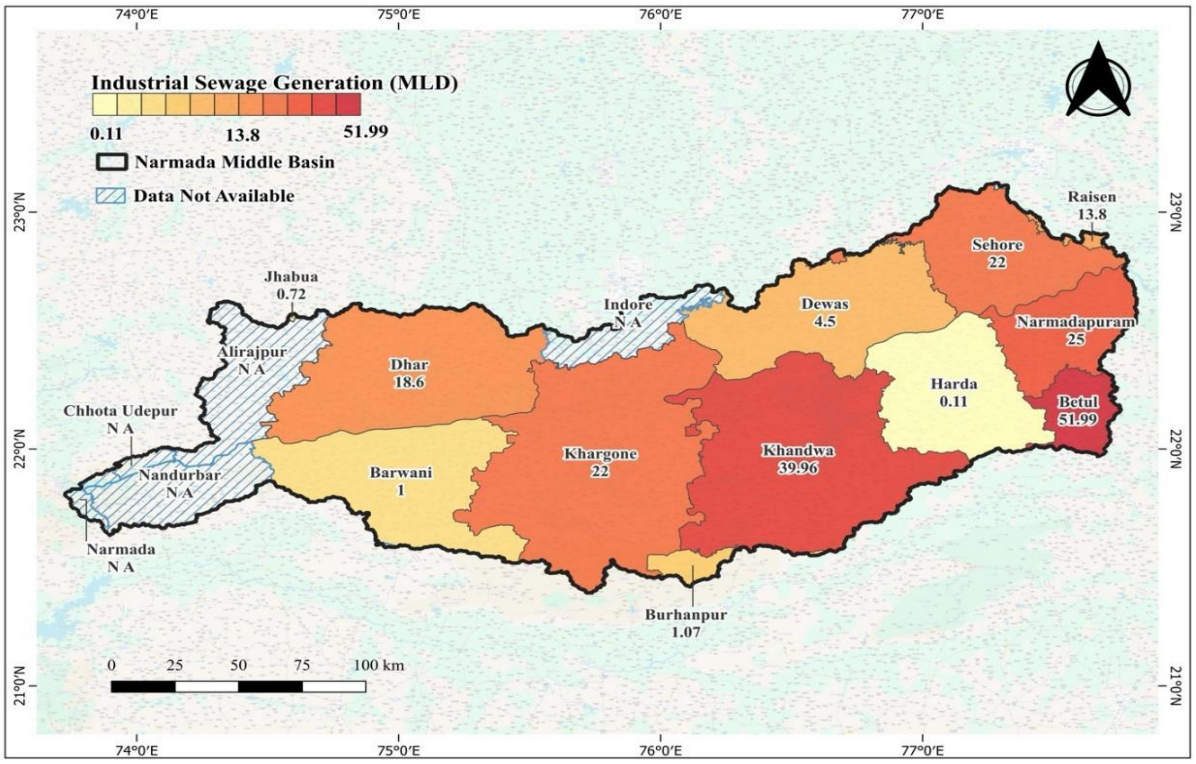


Figure 8 District-wise Industrial Sewage Generation (MLD) in Narmada Middle Basin

(Source: Madhya Pradesh Pollution Control Board (MPPCB). (2025). District Environmental Plans (DEPs))

4 Wastewater Treatment Facilities

4.1 Sewage Treatment Plants (STPs)

4.1.1 Upper Basin

4.1.1.1 Operational STPs

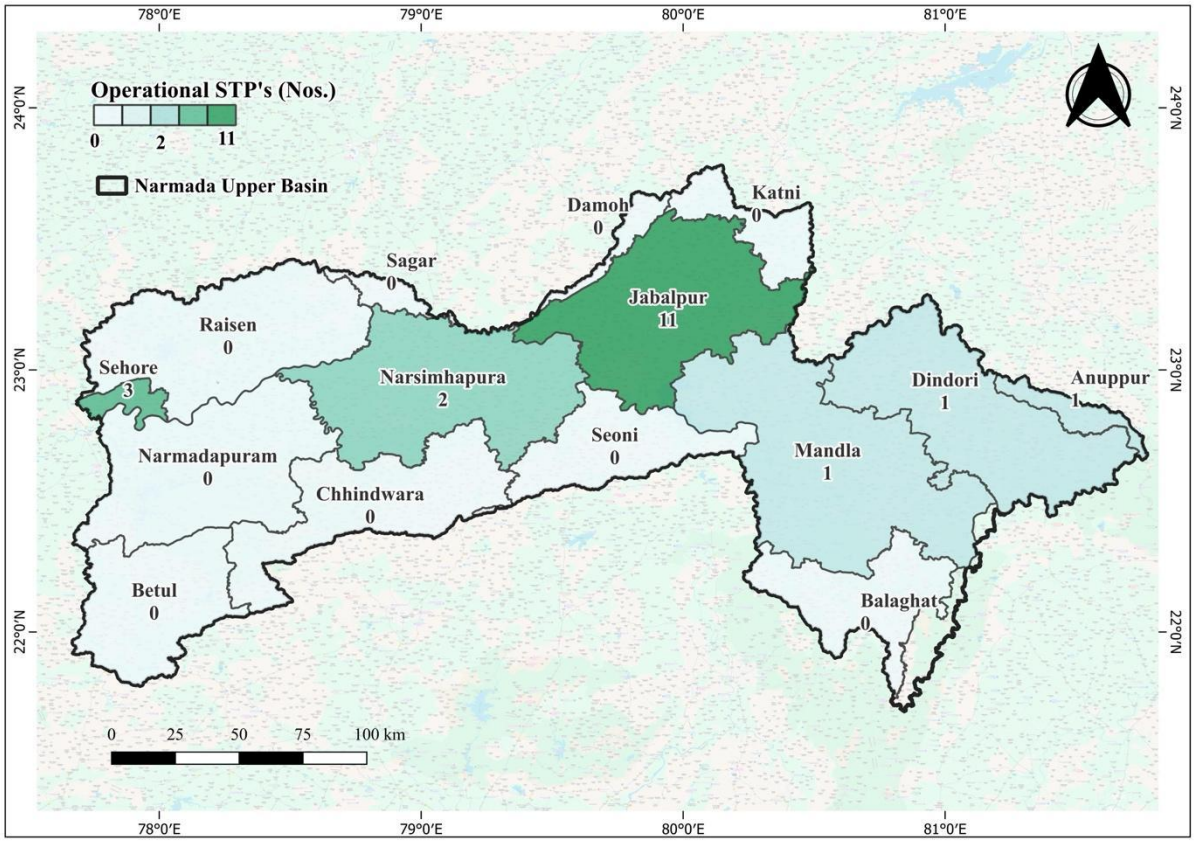


Figure 9 Number of Operational STPs per district in Narmada Upper Basin

(Source: Government of Madhya Pradesh. (2024). Action taken report in OA No. 606 of 2018 (Compliance of MSW Management Rules, 2016 and other environmental issues). National Green Tribunal.)

The Map shown in Fig. 9 provides a district-wise overview of the number of operational Sewage Treatment Plants (STPs) in the Narmada Upper Basin, Madhya Pradesh, highlighting regions with established sewage management infrastructures.

Jabalpur is the clear hotspot, hosting 11 operational STPs, the highest number in the basin. This significant concentration underscores Jabalpur's central role in regional wastewater treatment, likely reflecting a combination of large urban population, industrial presence, and strong administrative prioritization of sanitation efforts. Sehore (3 STPs) and Narsimhapura (2 STPs)

form secondary hotspots. Their multiple operational facilities enable more decentralized, reliable sewage management, benefiting local communities with improved environmental and public health conditions. Districts such as Mandla (1 STP) and Dindori (1 STP) have begun to establish sewage treatment capacity but lag in coverage. Anuppur (1 STP) and Balaghat (no number shown, presumed to be 0 or 1) show minimal STP infrastructure. The majority of districts including Narmadapuram, Betul, Raisen, Damoh, Katni, Chhindwara, Sagar, Seoni, and Balaghat currently have no operational STPs, indicated by the lightest map shades. These regions face ongoing challenges of untreated sewage, with risks of river pollution and health impacts.

The map highlights the need for targeted expansion of STP facilities in districts with either no or only one operational plant. Strengthening sewage infrastructure not only protects water quality in the Narmada Basin but also promotes sustainable urban growth, disease reduction, and ecological restoration.

4.1.1.2 STPs Under Construction

The Map shown in Fig. 10 illustrates the spatial distribution of under-construction Sewage Treatment Plants (STPs) across various districts of the Narmada Upper Basin. Districts such as Jabalpur, Mandla, Betul, and Sehore have the highest number of under-constructions STPs (2 each), highlighting their prioritization in wastewater treatment infrastructure. These areas are likely critical nodes for managing wastewater due to higher population densities or strategic river stretches. Districts including Raisen, Narmadapuram, Narsimhapur, Chhindwara, and Seoni each have 1 STP under construction, reflecting moderate investment in wastewater management infrastructure. Some districts, such as Dindori, Anuppur, Katni, and Damoh, currently show 0 STPs under construction, indicating either limited sewage management needs or projects still in the planning phase. This visualization is important for environmental planning and water quality management as it demonstrates ongoing efforts to strengthen sewage treatment infrastructure in the upper catchment of the Narmada River, which plays a crucial role in reducing pollution loads and enhancing river health.

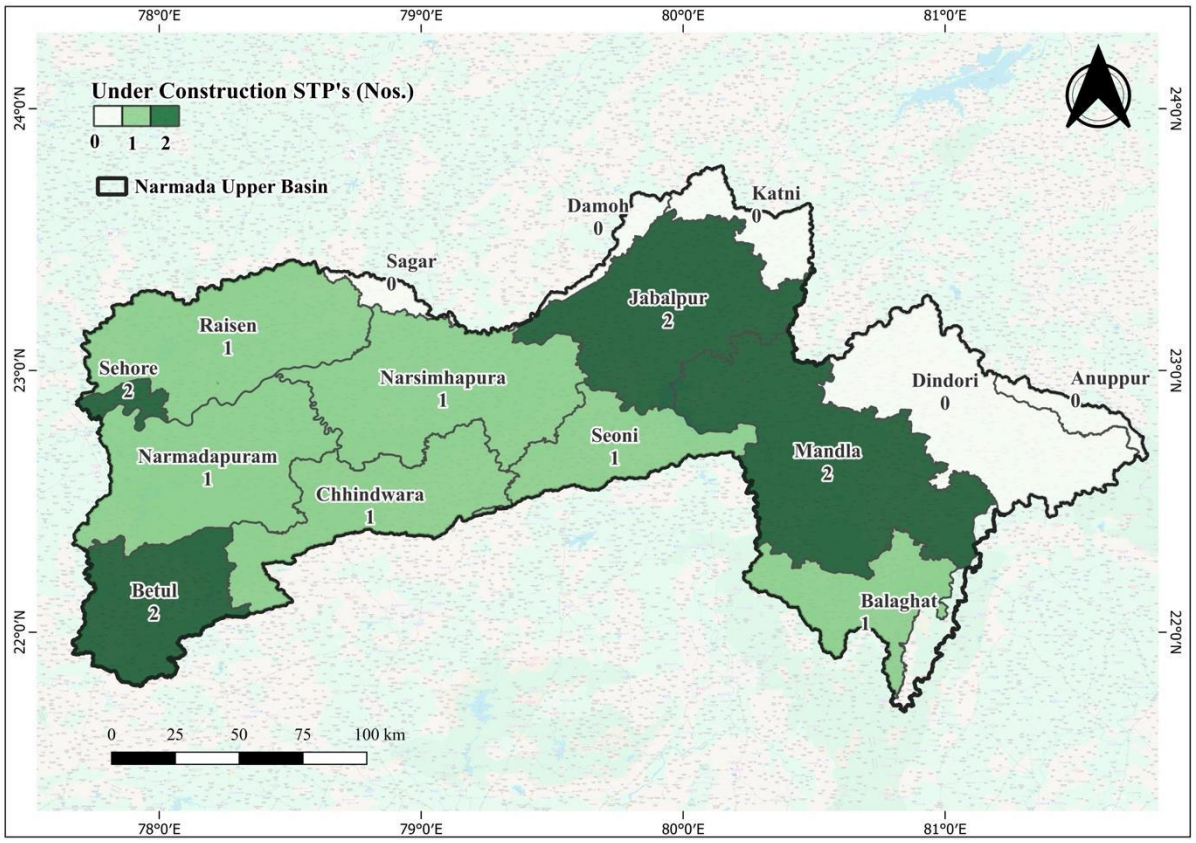


Figure 10 Number of Under Construction STPs per district in Narmada Upper Basin

(Source: Government of Madhya Pradesh. (2024). Action taken report in OA No. 606 of 2018 (Compliance of MSW Management Rules, 2016 and other environmental issues). National Green Tribunal.)

4.1.1.3 Cumulative Capacity of Operational STPs

The Map shown in Fig. 11 highlights operational sewage treatment across the Narmada Upper Basin districts in Madhya Pradesh, providing a clear spatial picture of where sewage is actively treated and where gaps persist.

The city of Jabalpur is the dominant hotspot, with an impressive operational STP capacity of 154.15 MLD, by far the largest in the basin. Mandla (17.6 MLD) and Sehore (15.28 MLD) form secondary hotspots, each showing substantial investment in functional sewage treatment infrastructure. These blue-shaded operational clusters represent centers where urban populations benefit from active wastewater cleansing, reduced river pollution, and improved public health outcomes. Narsimhapura (9 MLD), Dindori (3.85 MLD), and Anuppur (1.2 MLD) constitute smaller but notable hotspots. These districts are making tangible progress towards modern sanitation, directly benefiting both their local environments and downstream quality in the Narmada system. In contrast, districts such as Narmadapuram, Betul, Raisen,

Damoh, Katni, Chhindwara, Sagar, Seoni, and Balaghat currently lack operational STPs and appear as coldspots on the map. These areas face ongoing challenges regarding untreated sewage, with higher risk of waterborne diseases and pollution in natural water bodies. Addressing these gaps is vital for equitable water management across the Narmada basin.

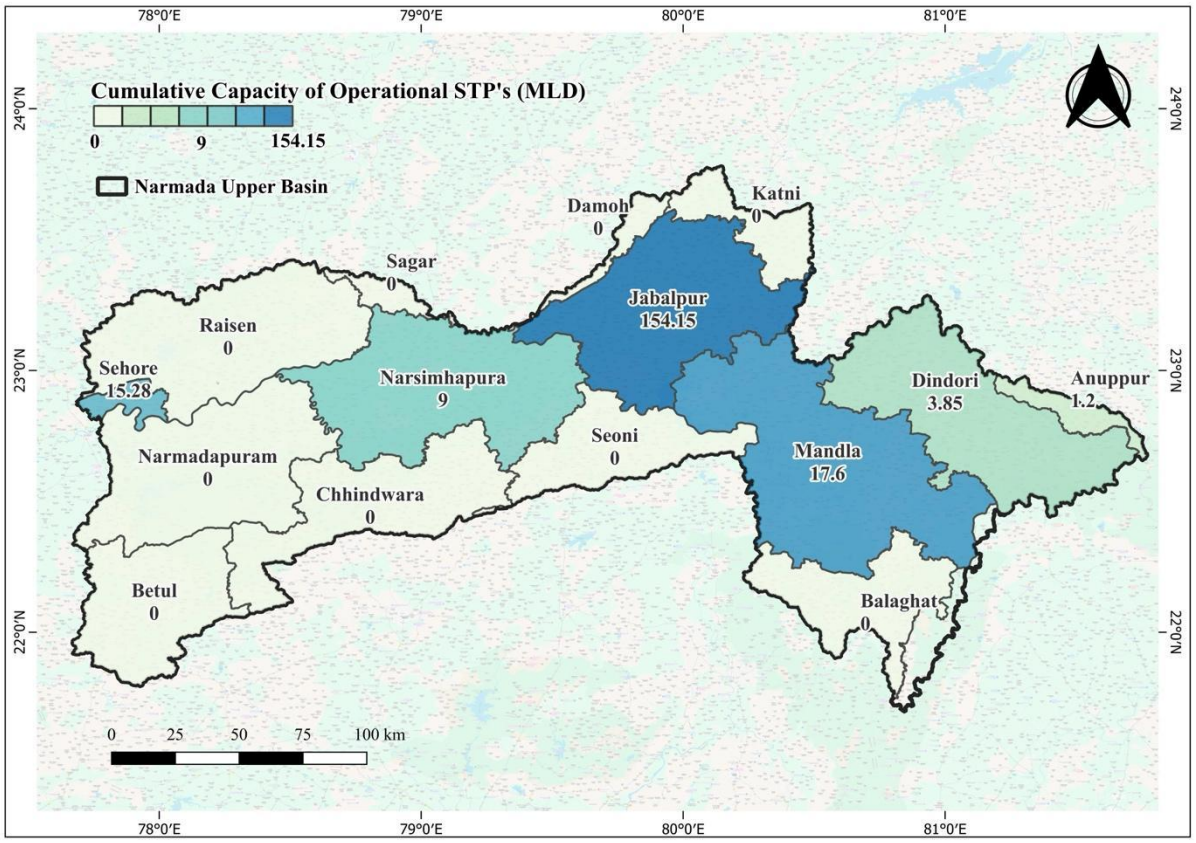


Figure 11 District-wise cumulative capacity (MLD) of Operational STPs in Narmada Upper Basin

(Source: Government of Madhya Pradesh. (2024). Action taken report in OA No. 606 of 2018 (Compliance of MSW Management Rules, 2016 and other environmental issues). National Green Tribunal.)

4.1.1.4 Cumulative Capacity of Under-Construction STPs in Upper Basin

The Map shown in Fig. 12 depicts the cumulative capacity of under-construction Sewage Treatment Plants (STPs) across districts in the Narmada Upper Basin, Madhya Pradesh, India, measured in Million Liters per Day (MLD), providing insight into the spatial progress of wastewater infrastructure development.

Narmadapuram is the leading district for ongoing STP infrastructure, with a cumulative under-construction capacity of 21 MLD the highest on the map. Mandla (9.5 MLD) and Jabalpur (8.8 MLD) are also significant contributors, marked by darker green. These districts are proactively

expanding their wastewater management capabilities, which aligns with both high sewage generation volumes and priority for stricter pollution control to preserve basin water quality. Sehore (5.2 MLD), Chhindwara (3 MLD), Raisen (2.7 MLD), and Balaghat (1.5 MLD) sit within the moderate range. These areas are investing consistently in STP capacity, supporting balanced urban development, and enhancing regional resilience against untreated sewage discharge. Betul, Narsimhapura, and Seoni each have 1 MLD under construction, representing initial steps to upgrade sanitation infrastructure. Several districts Damoh, Katni, Sagar, Dindori, and Anuppur report zero under-construction STP capacity at present, indicated by the lightest shades. The lack of new projects in these areas may reflect lower sewage loads, budgetary constraints, or a need for greater prioritization in future planning cycles.

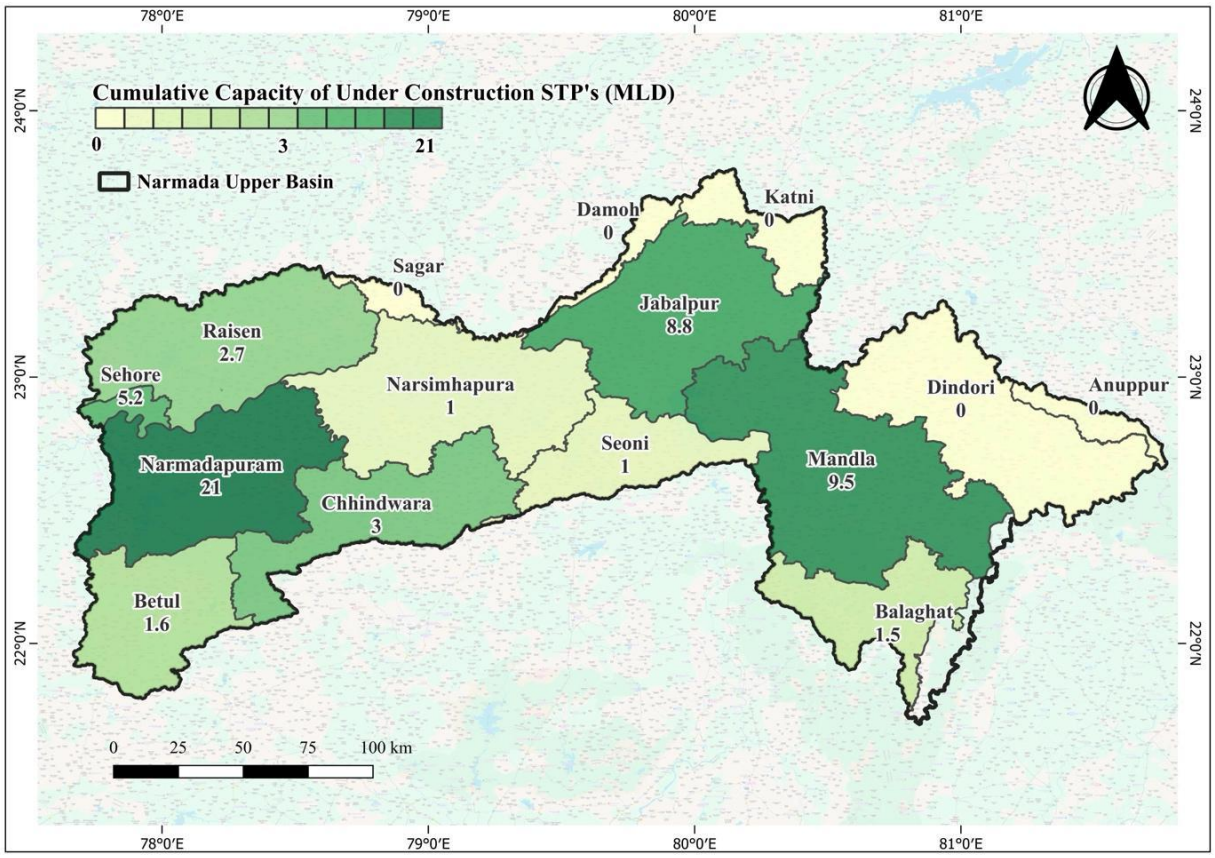


Figure 12 District-wise cumulative capacity (MLD) of Under Construction STPs in Narmada Upper Basin

(Source: Government of Madhya Pradesh. (2024). Action taken report in OA No. 606 of 2018 (Compliance of MSW Management Rules, 2016 and other environmental issues). National Green Tribunal.)

Narmadapuram, Mandla, and Jabalpur are key districts with robust STP capacity under construction, forming the backbone of future wastewater management in the basin. Moderate investment zones such as Sehore, Chhindwara, Raisen, Betul, Balaghat, Narsimhapura, and

Seoni reveal transitional progress toward improved sanitation coverage. Districts with zero capacity underway (Damoh, Katni, Sagar, Dindori, and Anuppur) need renewed focus in infrastructure planning, especially as urbanization accelerates. The map offers strategic regional intelligence for optimizing STP project rollout and tuning basin-level sustainability objectives.

4.1.2 Middle Basin

4.1.2.1 Operational STPs in Middle Basin

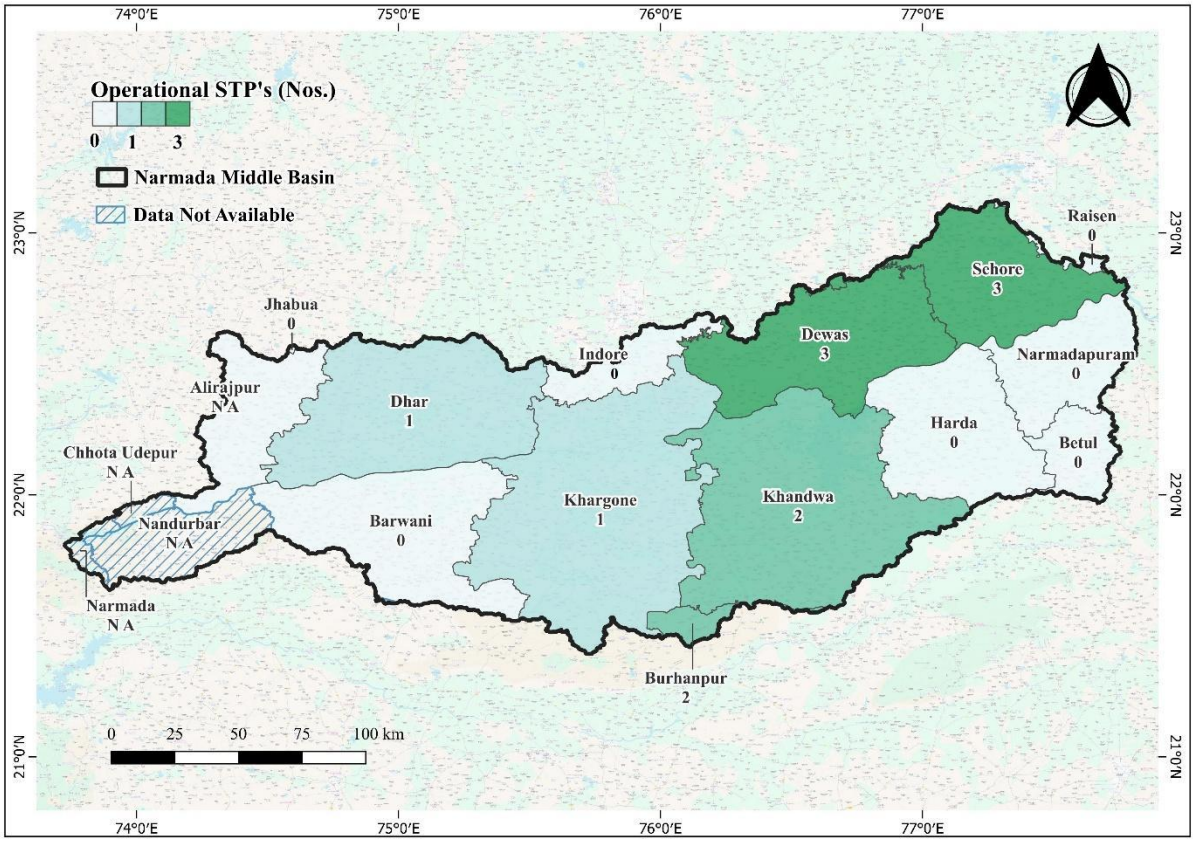


Figure 13 District-wise Operational STPs in Narmada Middle Basin

(Source: Government of Madhya Pradesh. (2024). Action taken report in OA No. 606 of 2018 (Compliance of MSW Management Rules, 2016 and other environmental issues). National Green Tribunal.)

The Map shown in Fig. 13 displays the distribution of operational Sewage Treatment Plants (STPs) by number across the districts of the Narmada Upper Basin. It highlights areas of significant infrastructure development and points out regions needing future attention. Jabalpur is the principal hotspot, housing 11 operational STPs, the highest number among all districts in the basin. This strong infrastructure reflects an advanced stage of urban sanitation

management, supporting Jabalpur's ability to process large sewage volumes from both residential and industrial sources. Jabalpur's leadership sets an example for regional urban centers regarding the expansion of wastewater treatment facilities. Sehore (3 STPs) and Narsimhapura (2 STPs) are notable secondary hotspots. Their increased number of plants suggests growing urban populations and progressive local governance with a focus on sustainable waste management. Many districts, including Narmadapuram, Betul, Raisen, Damoh, Katni, Chhindwara, Sagar, Seoni, and Balaghat, lack operational STPs altogether, or only have a single plant (such as Mandla, Dindori, and Anuppur). These light-shaded zones represent underserved regions where sewage treatment infrastructure is either nascent or absent, exposing local populations and waterways to greater health and pollution risks.

Jabalpur is the chief operational hotspot for STPs, setting the infrastructural standard for the Narmada Upper Basin. Sehore and Narsimhapura are emerging secondary clusters showing forward momentum in sewage management. Many districts require focused policy, planning, and investment to bridge large infrastructure gaps and promote sustainable sanitation outcomes across the basin.

4.1.2.2 Under Construction STPs in Middle Basin

The Map shown in Fig. 14 shows the distribution of under-construction Sewage Treatment Plants (STPs) by number across the districts of the Narmada Middle Basin. It highlights zones of present infrastructure growth and regions that require prioritization for future sewage management. Barwani and Khargone each have 4 under-construction STPs, emerging as the core hotspots for upcoming sewage treatment capacity in the basin. This active expansion reflects strong recognition of sanitation needs, the pressure of rapid urbanization, and responsive district-level planning to manage future wastewater loads. Dhar and Alirajpur each feature 2 plants under construction, signifying moderate levels of imminent infrastructure development. Sehore also joins the group of moderately active districts with 2 new facilities underway, positioning it for improved management of urban and peri-urban sewage. Several districts, including Raisen, Narmadapuram, Betul, and Burhanpur, report just 1 STP under construction, reflecting initial steps toward improved sewage treatment but leaving considerable coverage gaps. Many districts, such as Dewas, Harda, Khandwa, Indore, and others, have no current projects, as portrayed by the lightest coloration, indicating significant vulnerability to unmanaged wastewater issues as populations grow.

Several territorial units, namely Nandurbar, Narmada, and Chhota Udepur, are shown with crosshatch shading, representing areas where data is not available. These pose challenges for monitoring and infrastructure planning, and transparency improvements are needed.

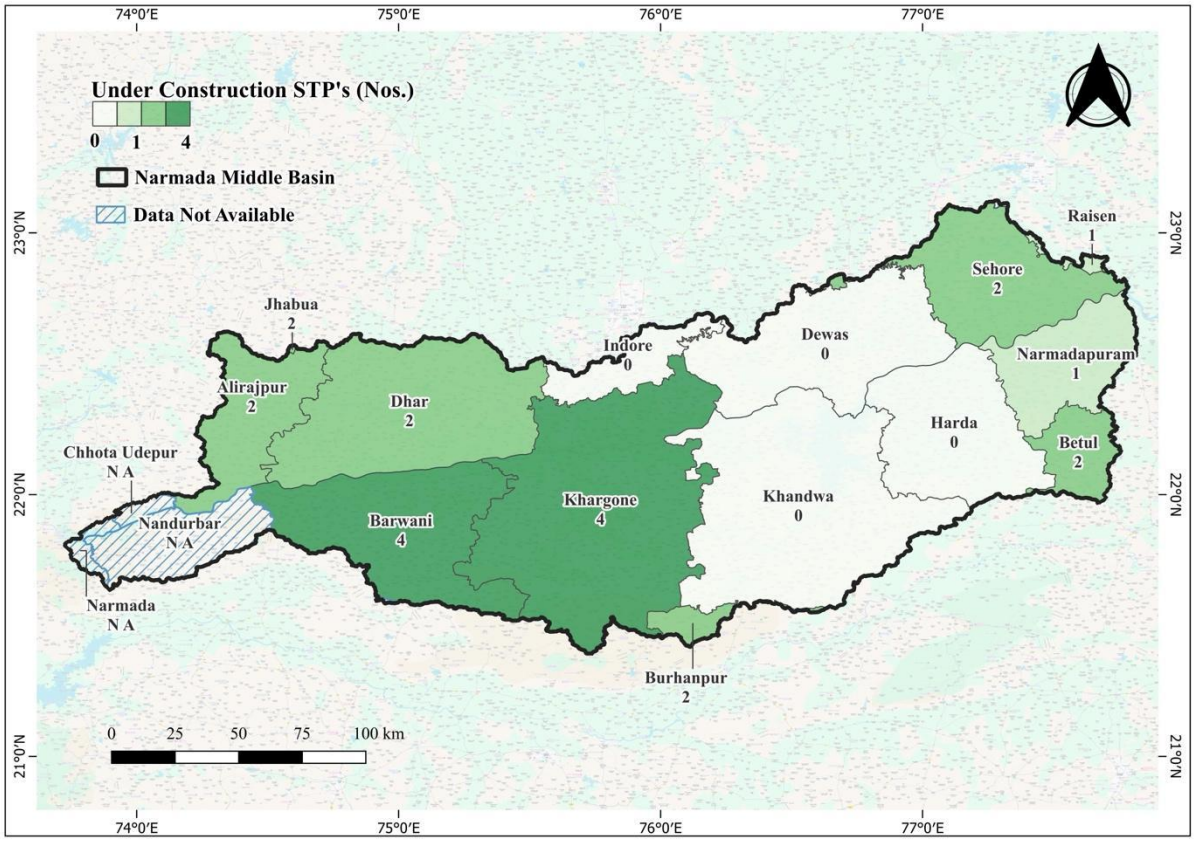


Figure 14 District-wise Under-Construction STPs in Narmada Middle Basin
(Source: Government of Madhya Pradesh. (2024). Action taken report in OA No. 606 of 2018 (Compliance of MSW Management Rules, 2016 and other environmental issues). National Green Tribunal.)

4.1.2.3 Cumulative Capacity of Operational STPs in Middle Basin

The Map shown in Fig. 15 illustrates the cumulative capacity of operational sewage treatment plants (STPs) across the districts of the Narmada Middle Basin, expressed in Million Liters per Day (MLD). It highlights which regions are already equipped to treat large amounts of sewage, and which continue to face infrastructure gaps.

Dewas (38 MLD) and Burhanpur (24 MLD) are the leading hotspots, shown with the darkest blue, indicating strong operational infrastructure for sewage treatment. These districts are well-positioned to manage urban and industrial sewage flows, offering significant protection to local

water bodies and helping maintain sanitation standards for growing populations. Khandwa (17.95 MLD) and Khargone (17.6 MLD) also represent notable operational strengths, with considerable treatment capacities. Sehore (15.28 MLD) and Dhar (12 MLD) reinforce this emerging belt of sewage management readiness throughout the middle basin. Districts like Narsimhapura (not labeled, presumed moderate or minimal), along with lower-capacity districts such as Alirajpur (not labeled) and Indore (not labeled), tend to have less capacity but show progress in operational deployments, as reflected in lighter blue hues. Consistent investment and scaling in these areas will help address cumulative sanitation needs.

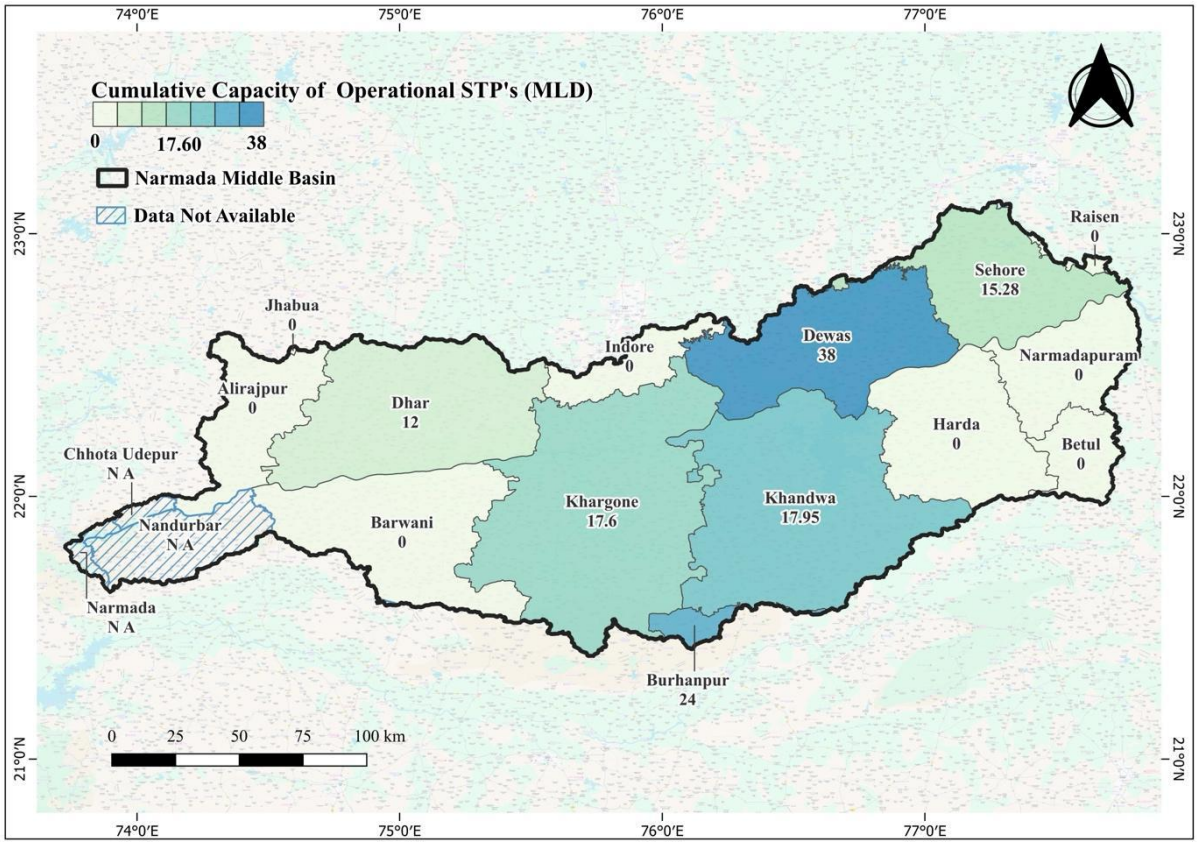


Figure 15 District-wise cumulative capacity (MLD) of Operational STPs in Narmada Middle Basin

(Source: Government of Madhya Pradesh. (2024). Action taken report in OA No. 606 of 2018 (Compliance of MSW Management Rules, 2016 and other environmental issues). National Green Tribunal.)

Several districts, such as Harda, Narmadapuram, Betul, Raisen, and Barwani, record zero operational STP capacity, highlighted in the lightest coloration. Chhota Udepur, Nandurbar, and Narmada are marked as data not available, which flags a transparency and monitoring gap.

These regions are the most vulnerable to risks from untreated wastewater—public health hazards, environmental contamination, and long-term river degradation.

4.1.2.4 Cumulative Capacity of Under-Construction STPs in the Middle Basin

The Map shown in Fig. 16 presents the cumulative capacity of under-construction Sewage Treatment Plants (STPs) across districts in the Narmada Middle Basin, measured in Million Liters per Day (MLD). It reveals which districts are rapidly upgrading their wastewater treatment infrastructure and which areas are yet to expand their capacity.

Narmadapuram (21 MLD), Barwani (20 MLD), Khargone (16 MLD), Dhar (11 MLD), Jhabua (8 MLD), and Alirajpur (4 MLD) emerge as key hotspots for ongoing STP development, illustrated by dark green shades. These districts are showing strong progress, signaling recognition of wastewater management needs and commitment to clean river basin practices. Significant under-construction capacity in these areas will enhance future sewage treatment coverage, reduce environmental pollution, and improve public health. Sehore (5 MLD), Raisen (3 MLD), Burhanpur (2 MLD), and Betul (1 MLD) demonstrate moderate investments in new STP capacity. These ongoing expansions will help support growing urban populations and prevent the escalation of untreated wastewater problems.

Several districts, including Indore, Dewas, Harda, and Khandwa, currently show zero under-construction STP capacity, indicated by the lightest coloring. Additionally, Nandurbar, Chhota Udepur, and Narmada have striped shading to denote that data is not available. These gaps may reflect delays in infrastructure planning, lower urbanization, or reporting limitations, but they represent priority areas for future intervention and improved monitoring.

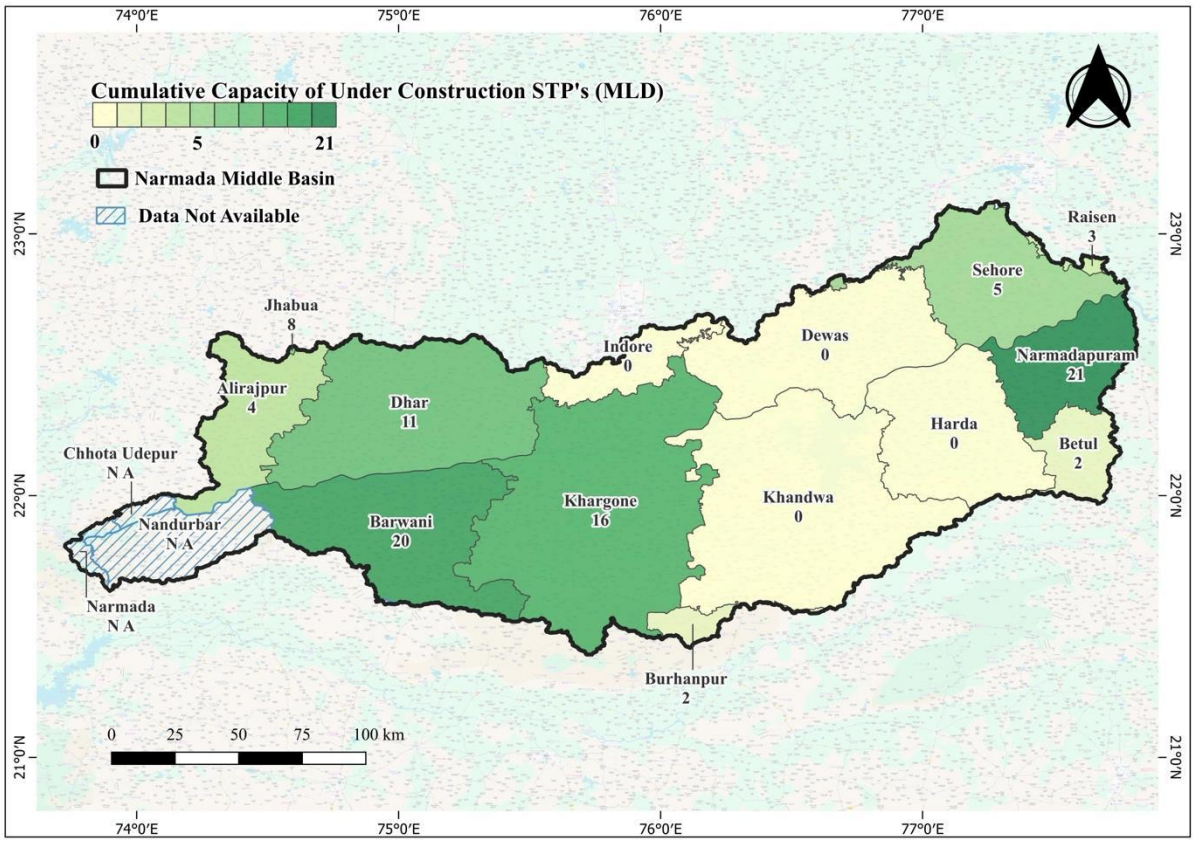


Figure 16 District-wise cumulative capacity (MLD) of under-construction STPs in Narmada Middle Basin

(Source: Government of Madhya Pradesh. (2024). Action taken report in OA No. 606 of 2018 (Compliance of MSW Management Rules, 2016 and other environmental issues). National Green Tribunal.)

5 Status of Solid Waste Generation in the Basin

5.1 Upper Basin

5.1.1 Solid Waste Generation in Upper Basin

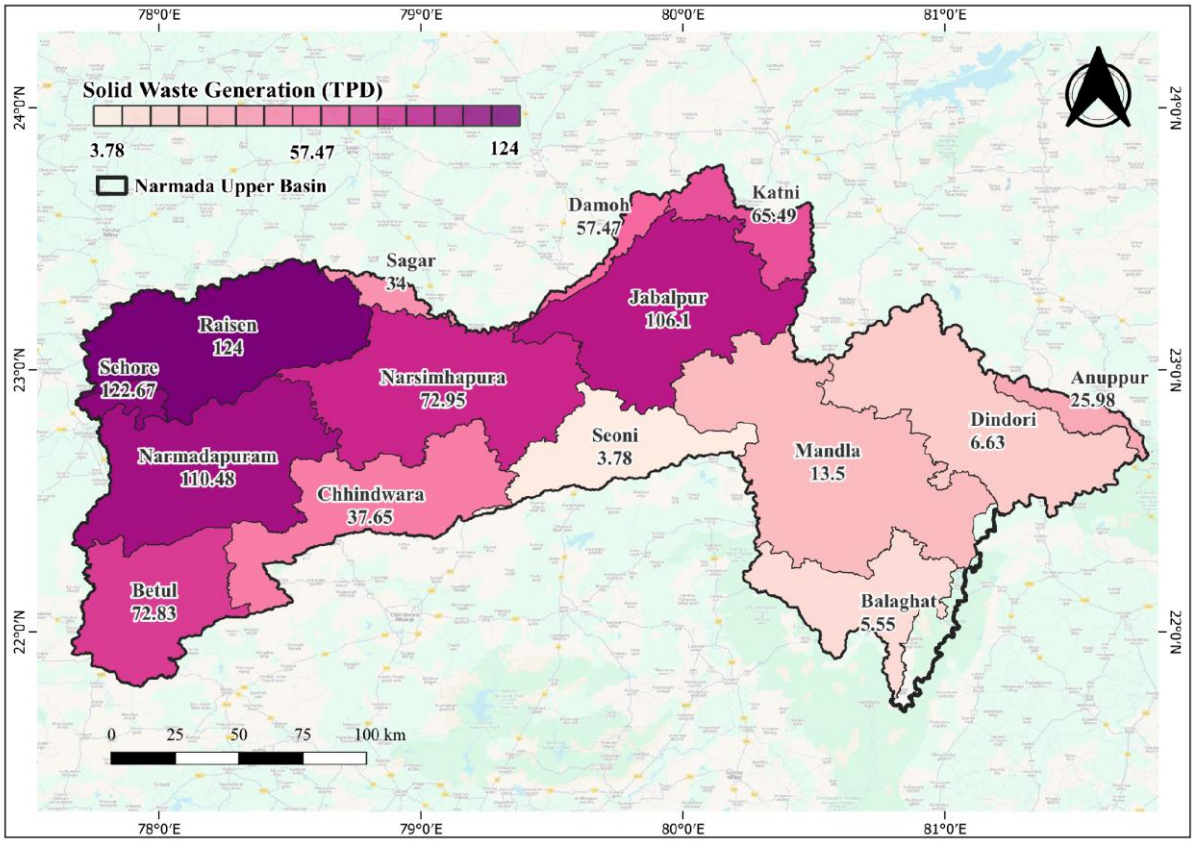


Figure 17 District-wise Solid Waste Generation (TPD) in the Narmada Upper Basin

(Source: Madhya Pradesh Pollution Control Board (MPPCB). (2025). District Environmental Plans (DEPs))

The Map shown in Fig. 17 identifies solid waste generation hotspots across districts in the Narmada Upper Basin, Madhya Pradesh, India, with a spatial focus on regions exhibiting high daily waste production, measured in Tons Per Day (TPD). The districts of Jabalpur (106.1 TPD), Sehore (122.67 TPD), Raisen (124 TPD), and Narmadapuram (110.48 TPD) emerge prominently as solid waste generation hotspots, represented by the deepest shades on the map. These urbanized zones indicate substantial solid waste management needs driven by dense populations, commercial activity, and rapid urban growth. Strategic intervention in these districts is critical for containing environmental pollution, promoting effective municipal waste management systems, and preventing the proliferation of legacy dumpsites.

Districts such as Katni (65.49 TPD), Damoh (57.47 TPD), Betul (72.83 TPD), and Narsimhapura (72.95 TPD) present moderate solid waste loads, serving as secondary hotspot zones. These areas require scaled localized solutions—such as waste segregation, decentralized composting, and robust collection infrastructure to reduce landfill dependence and address mounting daily waste volumes.

In contrast, districts like Seoni (3.78 TPD), Mandla (13.5 TPD), Balaghat (5.55 TPD), Dindori (6.63 TPD), and Anuppur (25.98 TPD) register lower values and are displayed in lighter tones. These peripheral regions, while presenting relatively modest waste management challenges, can benefit from early adoption of best practices, community engagement, and innovations in decentralized waste processing to prevent future escalation as populations grow.

5.2 Middle Basin

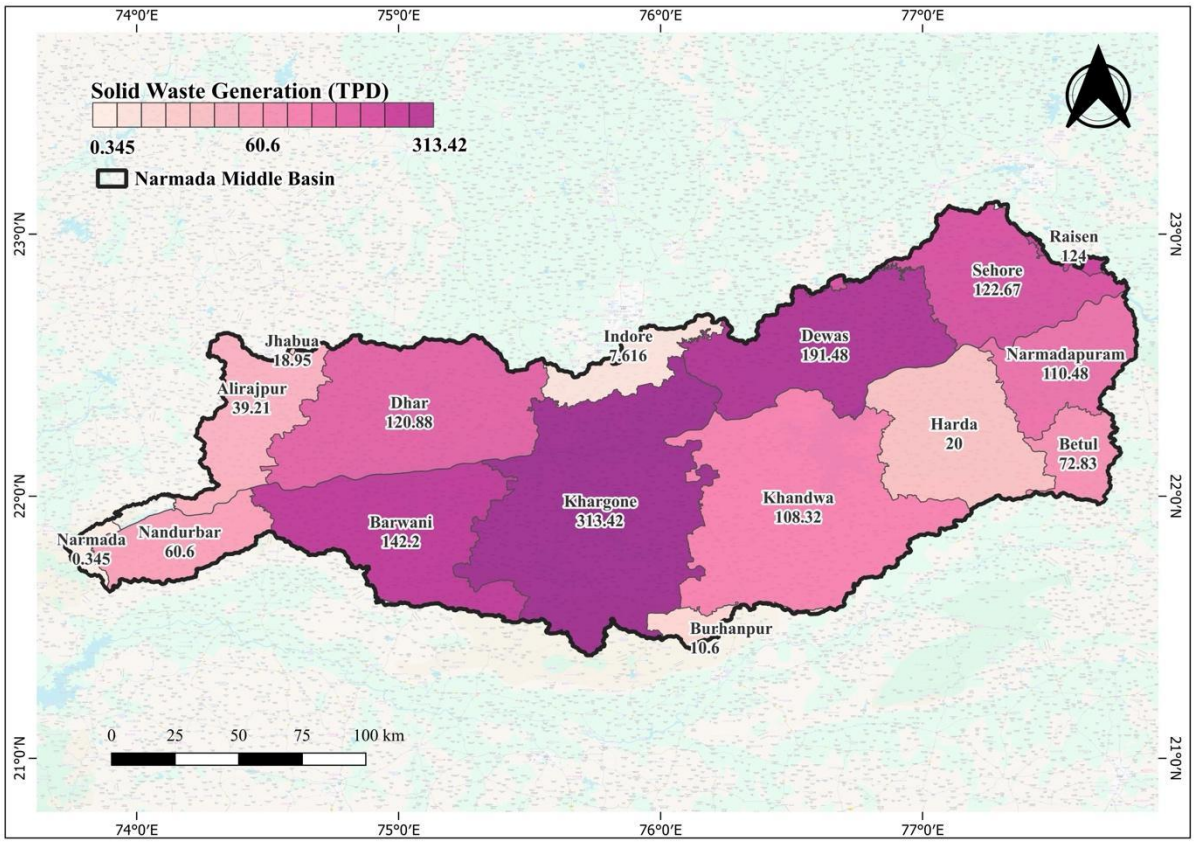


Figure 18District-wise Solid Waste Generation (TPD) in the Narmada Middle Basin

(Source: Madhya Pradesh Pollution Control Board (MPPCB). (2021,2025). District Environmental Plans (DEPs))

The Map shown in Fig. 18 visualizes daily solid waste generation (TPD) for each district in the Narmada Middle Basin, showcasing hotspot regions where municipal and commercial

activities generate the most waste. Khargone (313.42 TPD), Barwani (142.5 TPD), Dewas (191.48 TPD), and Sehore (122.67 TPD) are the principal hotspots, marked in the deepest purple. These districts have thriving urban centers, diverse economic activities, and large populations—leading to massive daily solid waste production. They urgently require advanced waste management systems such as modern landfill sites, segregation facilities, composting units, and public awareness campaigns about recycling and reduction. Narmadapuram (110.48 TPD), Khandwa (108.32 TPD), Dhar (120.88 TPD), Raisen (124 TPD), Betul (72.83 TPD), Burhanpur (10.6 TPD), Indore (76.16 TPD), and Alirajpur (39.21 TPD) show moderate waste generation. Their lighter shading signals steady urban expansion and emerging commercial hubs. These districts can benefit from scaling up decentralized waste management approaches, promoting structured door-to-door collection, and building community partnerships for sustainable waste practices. Districts such as Harda (20 TPD), Jhabua (18.95 TPD), Nandurbar (60.6 TPD), and Narmada (0.345 TPD) have the lowest daily generation rates. Rural or semi-rural characteristics explain their minimal loads. Nonetheless, foundational interventions in waste reduction, composting, and public education will help them prepare for future growth and avoid waste accumulation problems.

6 Status of Hazardous Waste Generation in the Basin

6.1 Upper Basin

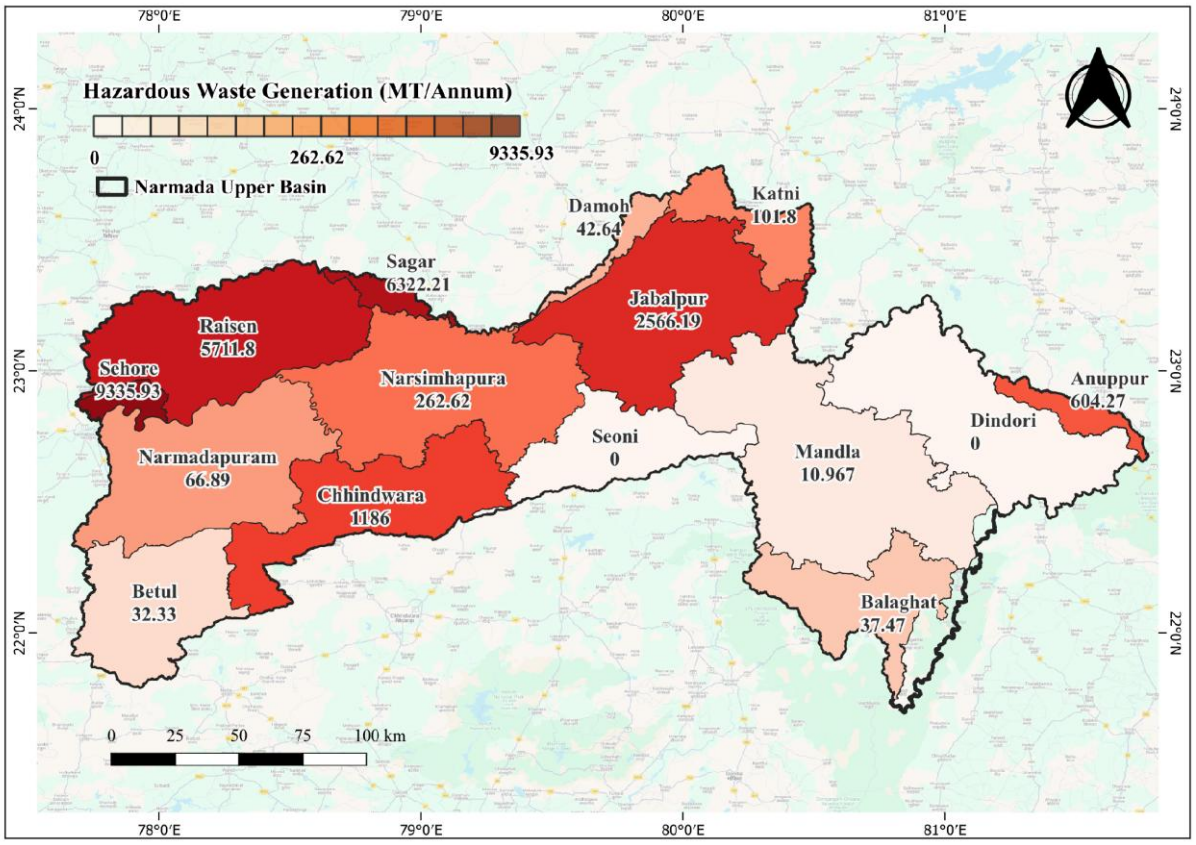


Figure 19 District-wise Hazardous Waste Generation (MT/Annum) in the Narmada Upper Basin

(Source: Madhya Pradesh Pollution Control Board (MPPCB). (2025). District Environmental Plans (DEPs))

The Map shown in Fig. 19 highlights the hazardous waste generation hotspots across the districts of the Narmada Upper Basin, Madhya Pradesh, India, with particular focus on areas producing the highest annual tonnage (MT/Annum) of hazardous waste. Sehore district stands out as the most significant hazardous waste hotspot, generating a massive 9335.93 MT per annum, followed by Sagar (6322.21 MT/annum), Raisen (5711.8 MT/annum), and Jabalpur (2566.19 MT/annum). The darkest map shades pinpoint these zones, indicating extensive industrial activity and waste-intensive processes, likely due to the presence of large-scale manufacturing, processing, or specialized waste-handling facilities. Several other districts Chhindwara (1186 MT/annum), Anuppur (604.27 MT/annum), Katni (101.18 MT/annum), and Narsimhapura (262.62 MT/annum) register moderate yet non-trivial quantities of hazardous

waste. These districts, found in intermediate shades, are essential focal points for regulatory intervention, infrastructure upgrades, and pollution surveillance to ensure containment and safe disposal of hazardous materials. Districts such as Mandla (10.967 MT/annum), Balaghat (37.47 MT/annum), Damoh (42.64 MT/annum), and Narmadapuram (66.89 MT/annum) display lower hazardous waste volumes, while some, notably Dindori and Seoni, report negligible or zero generation. These zones pose less immediate risk but must maintain vigilant environmental monitoring and preparedness as local industrialization patterns evolve.

6.2 Middle Basin

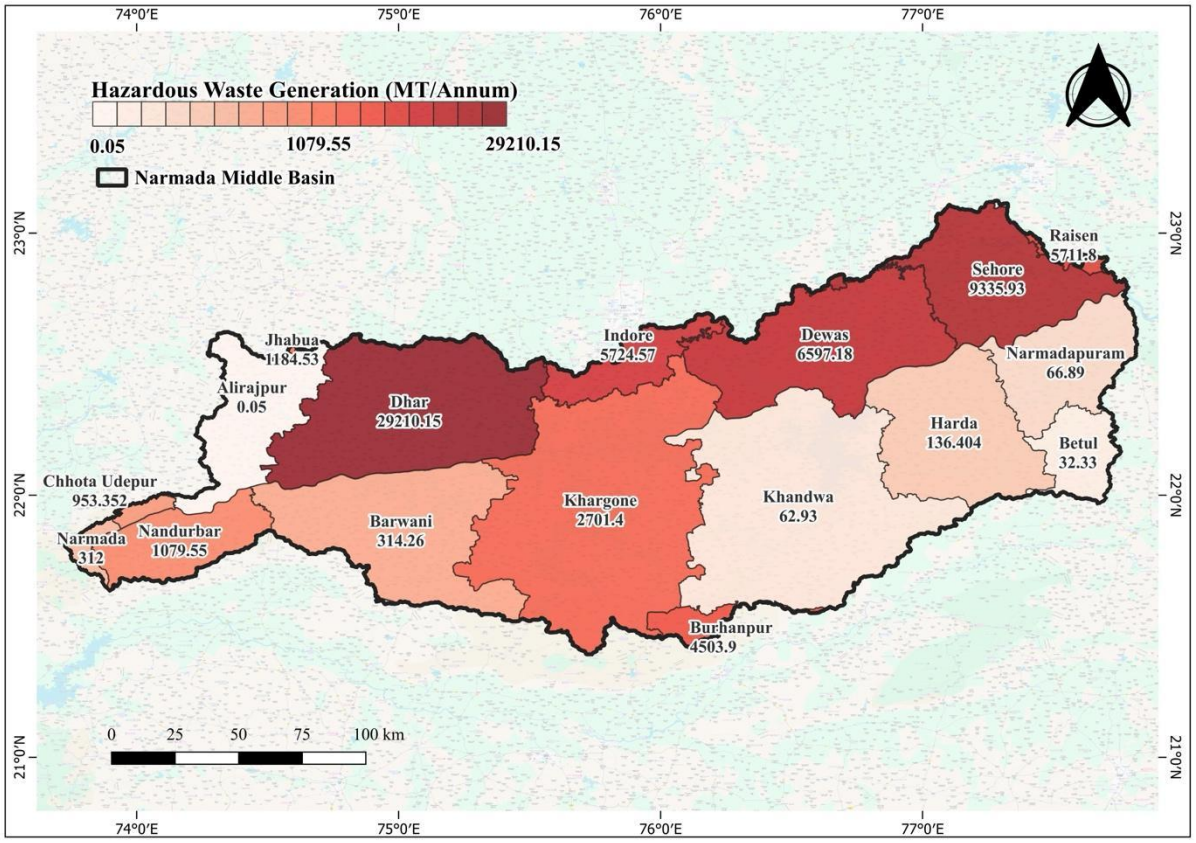


Figure 20District-wise Hazardous Waste Generation (MT/Annum) in the Narmada Middle Basin

(Source: Madhya Pradesh Pollution Control Board (MPPCB). (2025). District Environmental Plans (DEPs))

The Map shown in Fig. 20 visualizes hazardous waste generation (in MT/annum) across the districts of the Narmada Middle Basin, spotlighting key industrial and pollution hotspots.Dhar emerges as the most critical hotspot, generating a staggering 29,210.15 MT of hazardous waste annually. Sehore (9,335.93 MT), Indore (5,724.57 MT), Dewas (6,597.18 MT), and Burhanpur (4,503.9 MT) also stand out with some of the highest figures. These dark-shaded districts are

major industrial and urban centers, where intense manufacturing and processing activities create substantial volumes of hazardous by-products. The high concentration of such waste underscores the urgent need for robust waste management, strict regulatory oversight, and the adoption of best practices to reduce risks to people and the environment. Additional districts generating notable hazardous waste include Khargone (2,701.4 MT), Jhabua (1,184.53 MT), Barwani (314.26 MT), Nandurbar (1,079.55 MT), and Chhota Udepur (953.352 MT). While not as extreme as the primary hotspots, these areas still require vigilant waste monitoring, better treatment infrastructure, and continuous public health safeguarding. Some districts, such as Betul (32.33 MT), Khandwa (62.93 MT), Harda (136.404 MT), and Narmadapuram (66.89 MT), produce much smaller quantities, visualized in the lightest shades. Although their contribution to hazardous waste loads is comparatively minor, proactive management measures are still important for future-proofing these regions against the risks associated with industrialization and pollution.

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7 Status of Biomedical Waste Generation in the Basin

7.1 Upper Basin

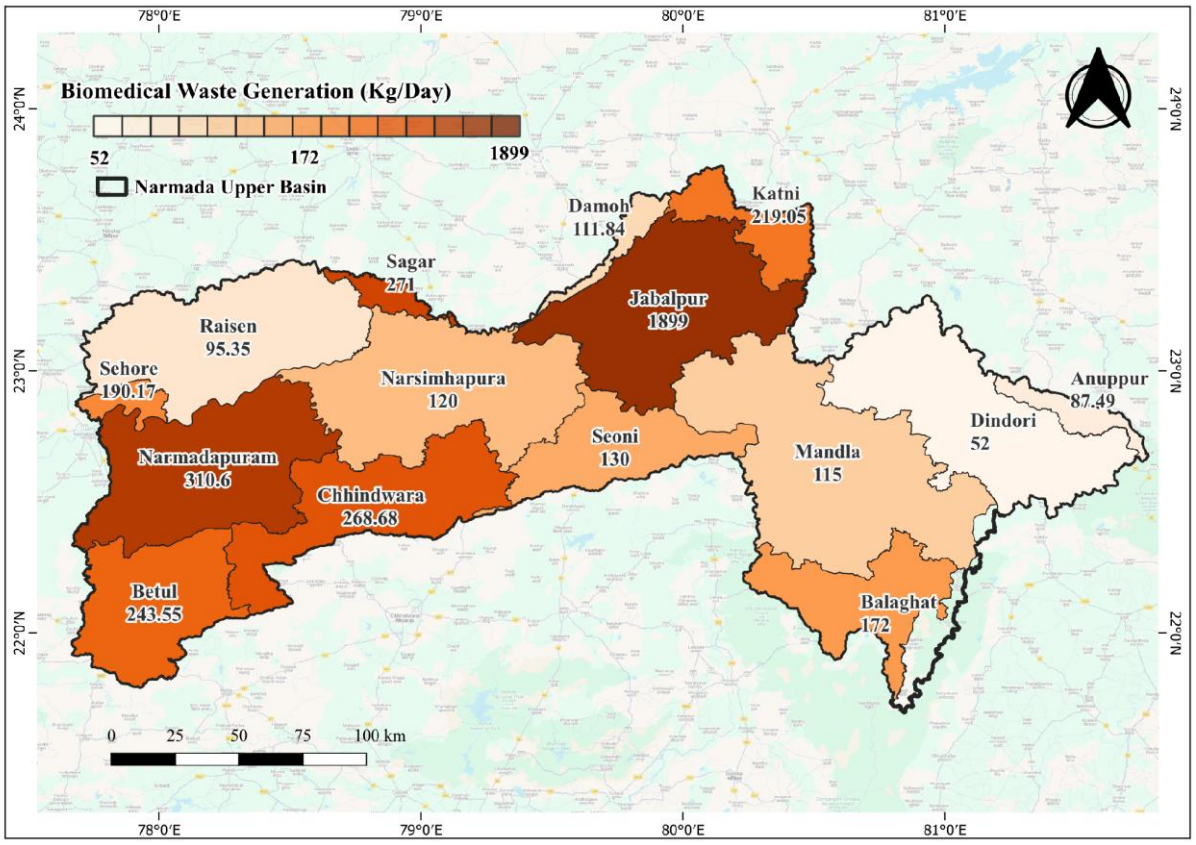


Figure 21 District-wise Biomedical Waste Generation (Kg/Day) in the Narmada Upper Basin

(Source: Madhya Pradesh Pollution Control Board (MPPCB). (2025). District Environmental Plans (DEPs))

The Map shown in Fig. 21 presents the spatial distribution of biomedical waste generation across districts in the Narmada Upper Basin, Madhya Pradesh, India, measured in kilograms per day (Kg/Day), highlighting regions of heightened biomedical waste burden linked to urban centers and healthcare facilities. Jabalpur (1899 Kg/Day) and Katni (2190.53 Kg/Day) are depicted as the most critical hotspots for biomedical waste generation, signaled by the darkest shades on the map. These districts harbor major urban centers, regional hospitals, and healthcare infrastructure, necessitating advanced biomedical waste segregation, treatment, and disposal systems to mitigate public health and environmental risks. Districts such as Narmadapuram (310.6 Kg/Day), Chhindwara (268.68 Kg/Day), Betul (243.55 Kg/Day), Sagar

(271 Kg/Day), and Sehore (190.17 Kg/Day) also generate considerable quantities of biomedical waste. While not at the extreme scale of Jabalpur or Katni, these areas represent significant secondary clusters requiring robust collection, incineration, and safe containment practices. The remaining districts—including Raisen (95.35 Kg/Day), Damoh (111.84 Kg/Day), Seoni (130 Kg/Day), Narsimhapura (120 Kg/Day), Mandla (115 Kg/Day), Balaghat (172 Kg/Day), Anuppur (87.49 Kg/Day), and Dindori (52 Kg/Day)—display moderate to low biomedical waste loads. In these territories, strengthening decentralized waste management, staff training, and compliance monitoring can prevent future escalations as healthcare demand grows.

7.2 Middle Basin

The Map shown in Fig. 22 represents daily biomedical waste generation (Kg/Day) across the districts of the Narmada Middle Basin, highlighting regions where management and containment of medical waste are especially crucial. Khargone (368.97 kg/day), Nandurbar (553 kg/day), and Jhabua (207.3 kg/day) are the most prominent hotspots for biomedical waste production, illustrated with the darkest orange on the map. These districts likely feature significant healthcare infrastructure, including hospitals and clinics, and serve as central nodes for medical services in the region. This necessitates robust biomedical waste management facilities to prevent environmental and public health hazards associated with improperly handled infectious and hazardous medical materials. Khandwa (297.47 kg/day), Dewas (273.33 kg/day), Burhanpur (260.62 kg/day), Narmadapuram (310.6 kg/day), Betul (243.55 kg/day), Dhar (201.1 kg/day), Barwani (184.59 kg/day), Alirajpur (190 kg/day), Sehore (190.17 kg/day), and Indore (29 kg/day) all contribute substantially to the biomedical waste load. While not as extreme as the primary hotspots, these districts require consistent attention with waste segregation, timely collection, and secure treatment systems. Districts such as Chhota Udepur (0.25 kg/day), Harda (85 kg/day), and Raisen (95.35 kg/day) produce lower quantities, shown in the lightest shades. Although risk intensity is lower, the presence of any untreated biomedical waste can still pose health and ecological risks, suggesting a need for foundational waste management frameworks and periodic monitoring.

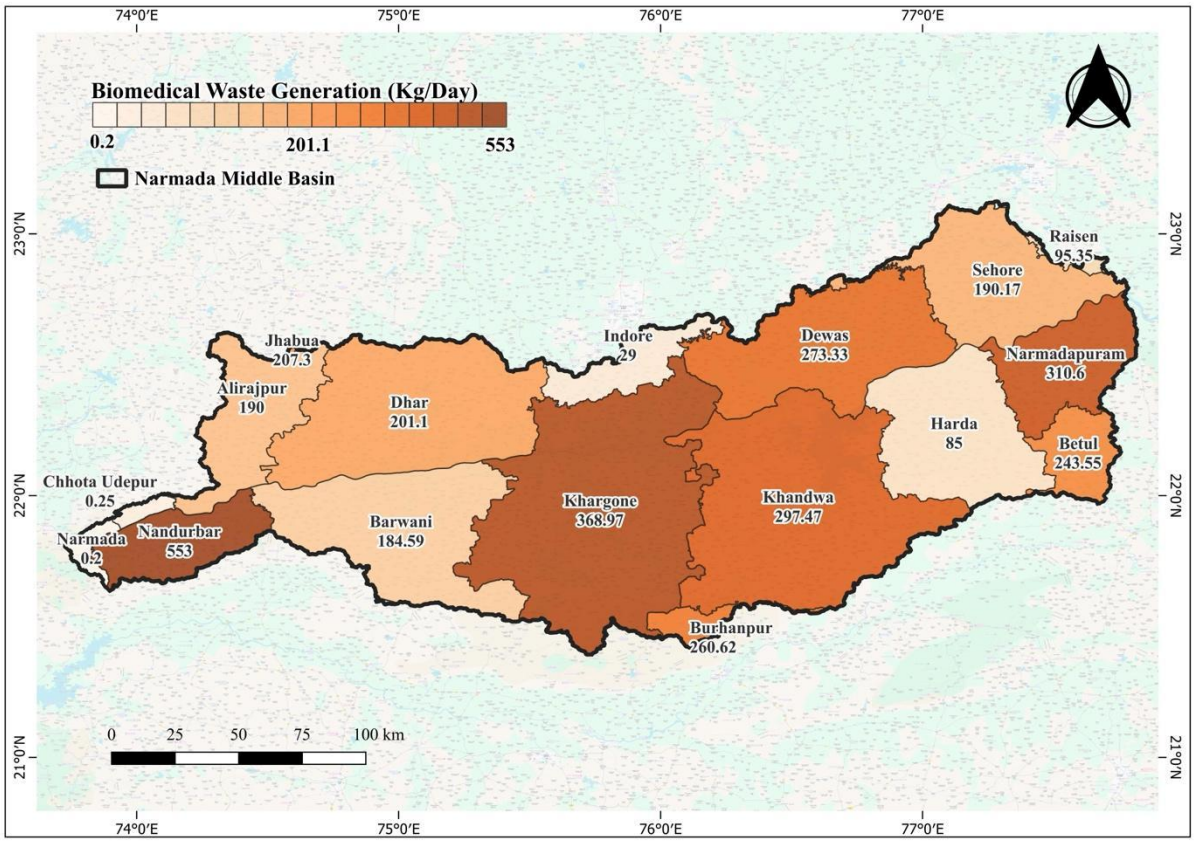


Figure 22 District-wise Biomedical Waste Generation (Kg/Day) in the Narmada Middle Basin

(Source: Madhya Pradesh Pollution Control Board (MPPCB). (2025). District Environmental Plans (DEPs))

Khargone, Nandurbar, and Jhabua are critical hotspots, demanding prioritized and sophisticated biomedical waste management solutions. Secondary hotspots must maintain steady increases in capacity and coverage as healthcare services expand. Peripheral regions should initiate best practices and ongoing monitoring to prevent risk escalation and safeguard local communities. The spatial analysis enables targeted actions for improved human and environmental safety across the Narmada Middle Basin. This map illustrates biomedical waste generation in the districts of the Narmada Middle Basin, with daily loads (in kg/day) visualized to highlight key hotspots where health infrastructure and population density drive substantial medical waste production.

8 Status of Plastic Waste Generation in the Basin

8.1 Upper Basin

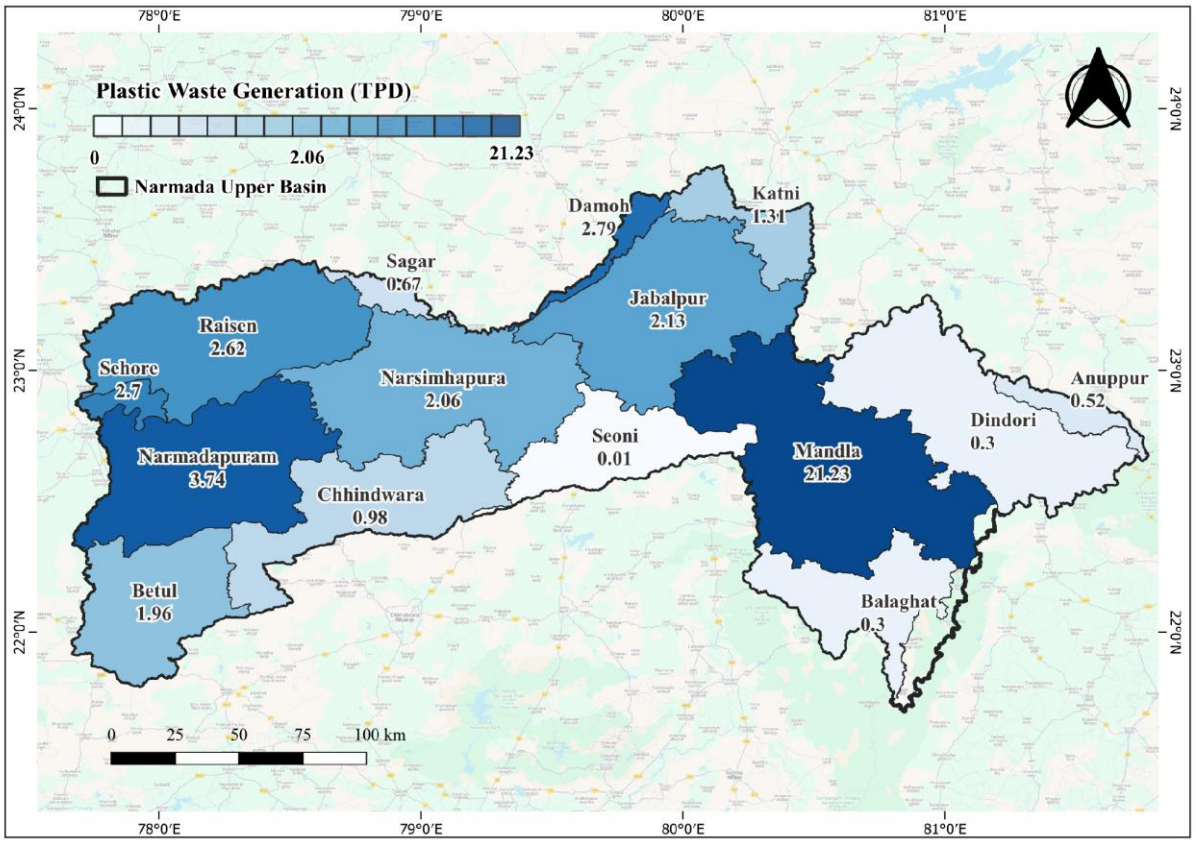


Figure 23 District-wise Plastic Waste Generation (TPD) in the Narmada Upper Basin

(Source: Madhya Pradesh Pollution Control Board (MPPCB). (2025). District Environmental Plans (DEPs))

The Map shown in Fig. 23 visualizes plastic waste generation across districts in the Narmada Upper Basin, Madhya Pradesh, India, measured in Tons Per Day (TPD), offering insight into the regions facing the most significant plastic waste management pressures.

Mandla district emerges as the dominant hotspot, generating 21.23 TPD of plastic waste marked by the darkest blue shade on the map. This exceptionally high figure suggests concentrated urban activity, prevalent market networks, and substantial consumer usage of plastic products, revealing an urgent need for targeted interventions in plastic collection, recycling, and reduction initiatives. Narmadapuram (3.74 TPD), Sehore (2.7 TPD), Damoh (2.79 TPD), Raisen (2.62 TPD), Jabalpur (2.13 TPD), Narsimhapura (2.06 TPD), and Betul (1.96 TPD) form a secondary ring of moderate plastic waste generators. These districts foster both urban and peri-urban settlements, indicating growing plastic use associated with

commercial activities and changing lifestyles. Focused awareness campaigns, improved segregation infrastructure, and extended producer responsibility (EPR) mechanisms are essential here. Other districts demonstrate comparatively minimal plastic waste output Katni (1.31 TPD), Chhindwara (0.98 TPD), Sagar (0.67 TPD), Anuppur (0.52 TPD), Dindori (0.3 TPD), Balaghat (0.3 TPD), and Seoni (0.01 TPD). These areas, painted in lighter shades, support limited urbanization and industrialization, but should still integrate preventive measures and scalable recycling infrastructure to address incremental rises as development proceeds. Mapping plastic waste hotspots aids regional authorities, municipal bodies, and community organizations in tailoring solutions. It prioritizes Mandla for immediate action, with secondary districts poised for scalable pilot programs. Participatory approaches such as public-private partnerships (PPP), waste pickers' empowerment, and implementation of plastic alternatives can amplify region-specific impact.

8.2 Middle Basin

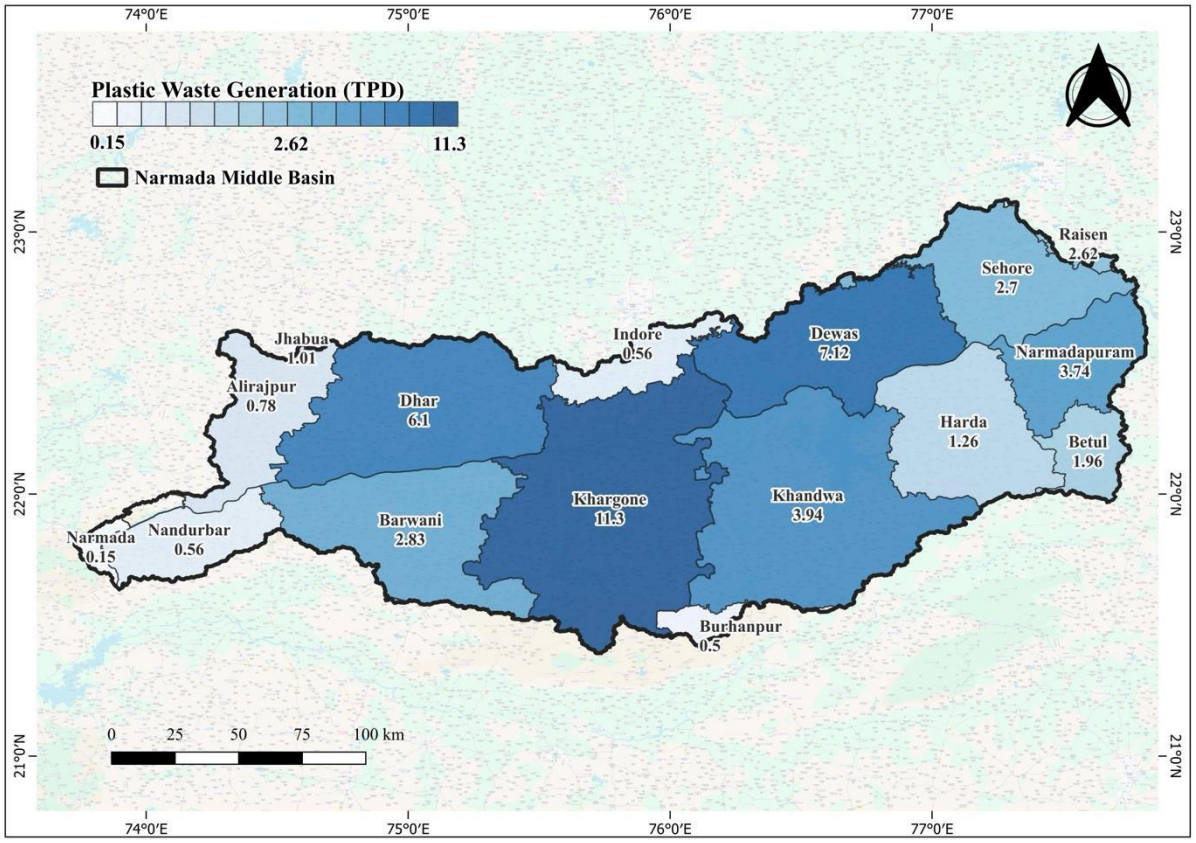


Figure 24 District-wise Plastic Waste Generation (TPD) in the Narmada Middle Basin

(Source: Madhya Pradesh Pollution Control Board (MPPCB). (2025). District Environmental Plans (DEPs))

The Map shown in Fig. 24 depicts the generation of plastic waste (in Tons Per Day) across the districts of the Narmada Middle Basin, highlighting areas with significant plastic pollution challenges and variation in plastic consumption patterns. Khargone (11.3 TPD) and Dewas (7.12 TPD) are the most pronounced plastic waste generation hotspots, indicated with the darkest blue shade. These districts have robust marketplaces, expanding urbanization, and high rates of packaged consumption, all contributing to larger volumes of discarded plastic. Addressing these loads requires investment in plastic collection systems, large-scale material recovery facilities, and public awareness drives targeting reduction and recycling. Dhar (6.1 TPD), Narmadapuram (3.74 TPD), Barwani (2.83 TPD), Khandwa (3.94 TPD), Raisen (2.62 TPD), Sehore (2.7 TPD), Betul (1.96 TPD), and Harda (1.26 TPD) generate moderate quantities. These areas reflect a blend of urban-rural transition and require a proactive approach strengthening door-to-door plastic collection, promoting alternatives, and supporting decentralized recycling initiatives. Burhanpur (0.5 TPD), Indore (0.56 TPD), Nandurbar (0.56 TPD), Alirajpur (0.78 TPD), Jhabua (1.01 TPD), and Narmada (0.15 TPD) fall in the lowest range, shown in the lightest blue. Their lower levels suggest a dominant rural character or effective waste minimization, but they should prioritize early adoption of anti-littering campaigns and small-scale recycling programs to prevent future accumulation.

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9 Status of C&D Waste Generation in the Basin

9.1 Upper Basin

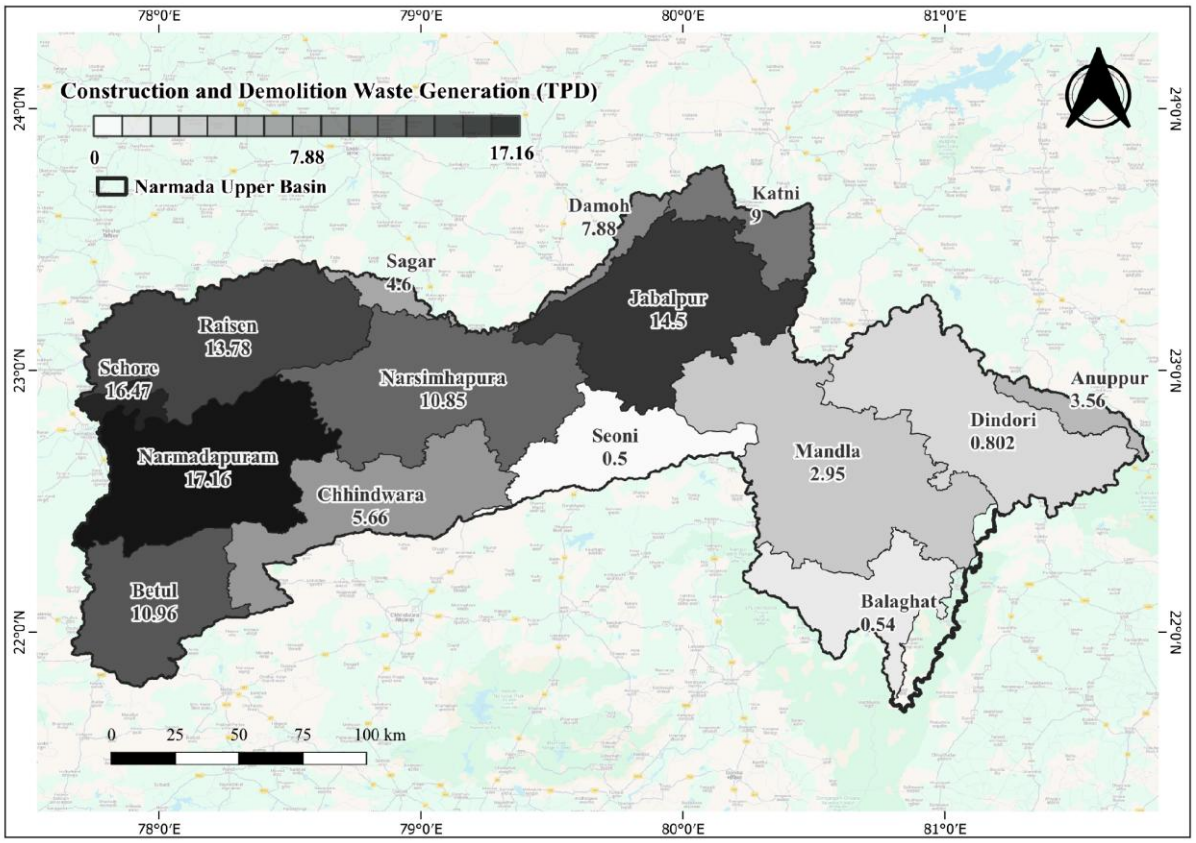


Figure 25 District-wise Construction and Demolition Waste Generation (TPD) in the Narmada Upper Basin, highlighting major and minor hotspots.

(Source: Madhya Pradesh Pollution Control Board (MPPCB). (2025). District Environmental Plans (DEPs))

The Map shown in Fig. 25 depicts the spatial distribution of construction and demolition (C&D) waste generation across the districts of the Narmada Upper Basin, Madhya Pradesh, India, with waste quantities expressed in Tons Per Day (TPD). The varying shades of gray illustrate the intensity of C&D waste generation, allowing identification of key hotspots. The western and central districts stand out as hotspots for C&D waste, with Narmadapuram (17.16 TPD), Sehore (16.47 TPD), Betul (10.96 TPD), Raisen (13.78 TPD), and Jabalpur (14.5 TPD) displaying the darkest tones on the map. These areas are marked by robust urban expansion, infrastructural activities, and significant real estate or industrial development. Such high levels of C&D waste require sustainable construction practices, efficient debris removal systems, and dedicated recycling facilities to reduce landfill pressure and support circular economy ambitions. Katni (9 TPD), Damoh (7.88 TPD), Narsimhapura (10.85 TPD), and Chhindwara

(5.66 TPD) form a secondary band of moderate waste generation, representing zones with advancing urbanization and incremental construction projects. These districts can benefit from targeted awareness programs on C&D waste minimization and incentivized recycling schemes. A majority of the eastern and southern districts, including Anuppur (3.56 TPD), Dindori (0.802 TPD), Balaghat (0.54 TPD), Mandla (2.95 TPD), and Seoni (0.5 TPD), are characterized by lighter shades, indicating low daily generation of C&D waste. These areas exhibit relatively limited construction activity, likely due to smaller urban footprints and less intensive infrastructure development.

9.2 Middle Basin

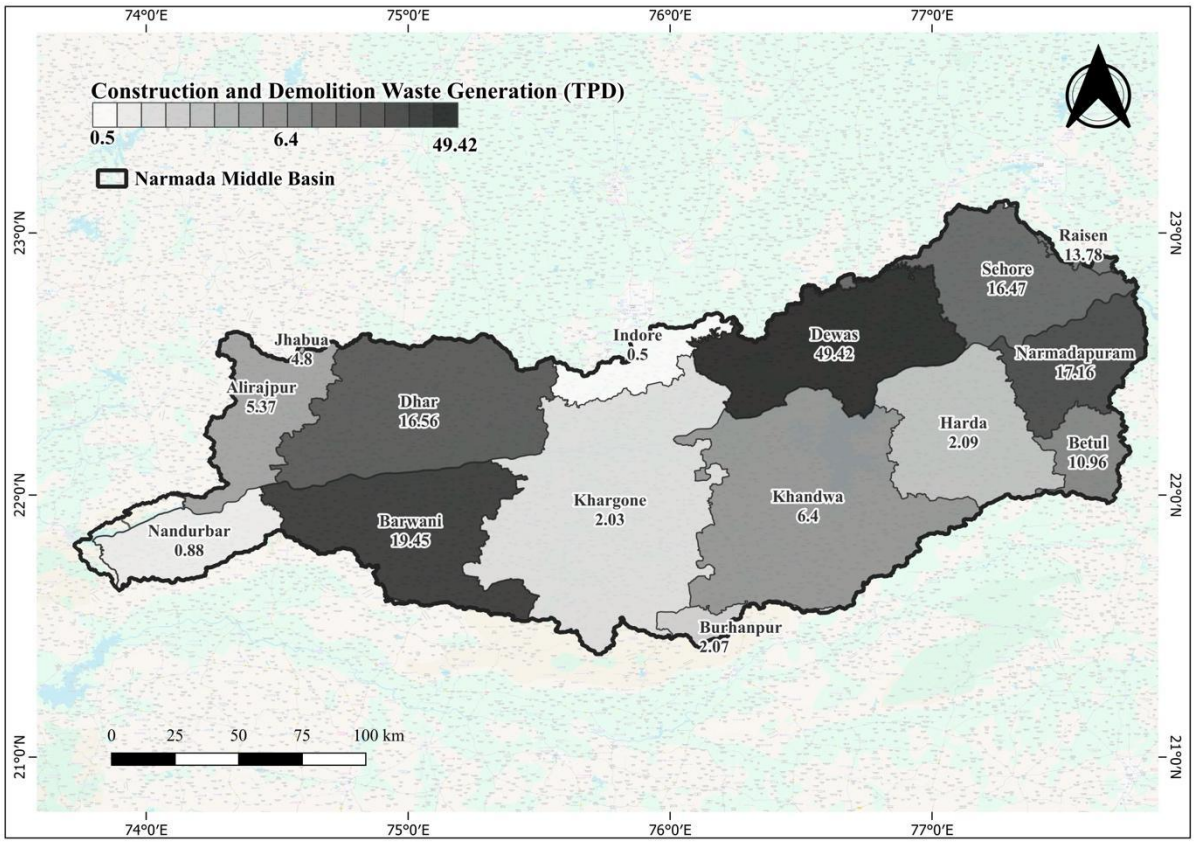


Figure 26 District-wise Construction and Demolition Waste Generation (TPD) in the Narmada Middle Basin, highlighting major and minor hotspots.

(Source: Madhya Pradesh Pollution Control Board (MPPCB). (2025). District Environmental Plans (DEPs))

The Map shown in Fig. 26 presents construction and demolition (C&D) waste generation across districts in the Narmada Middle Basin, quantified in tons per day (TPD). The distribution is visually represented in shades of gray, highlighting regions with higher and lower waste generation. Dewas district is the top contributor, generating 49.42 TPD, indicating significant

construction and demolition activities. Large volumes are also generated in Barwani (19.45 TPD), Sehore (16.47 TPD), Dhar (16.56 TPD), Narmadapuram (12.16 TPD), Raisen (13.78 TPD), and Betul (10.96 TPD), identifying these areas as construction hotspots within the basin. Districts like Khandwa (6.4 TPD), Alirajpur (5.37 TPD), Jhabua (4.8 TPD), Khargone (2.03 TPD), Harda (2.09 TPD), Burhanpur (2.07 TPD), Nandurbar (0.88 TPD), and Indore (0.5 TPD) represent zones with less intense C&D activity. The gradient from light to dark gray efficiently communicates areas of waste generation concern, aiding in prioritization for management interventions.

10 E-Waste

10.1 Upper Basin

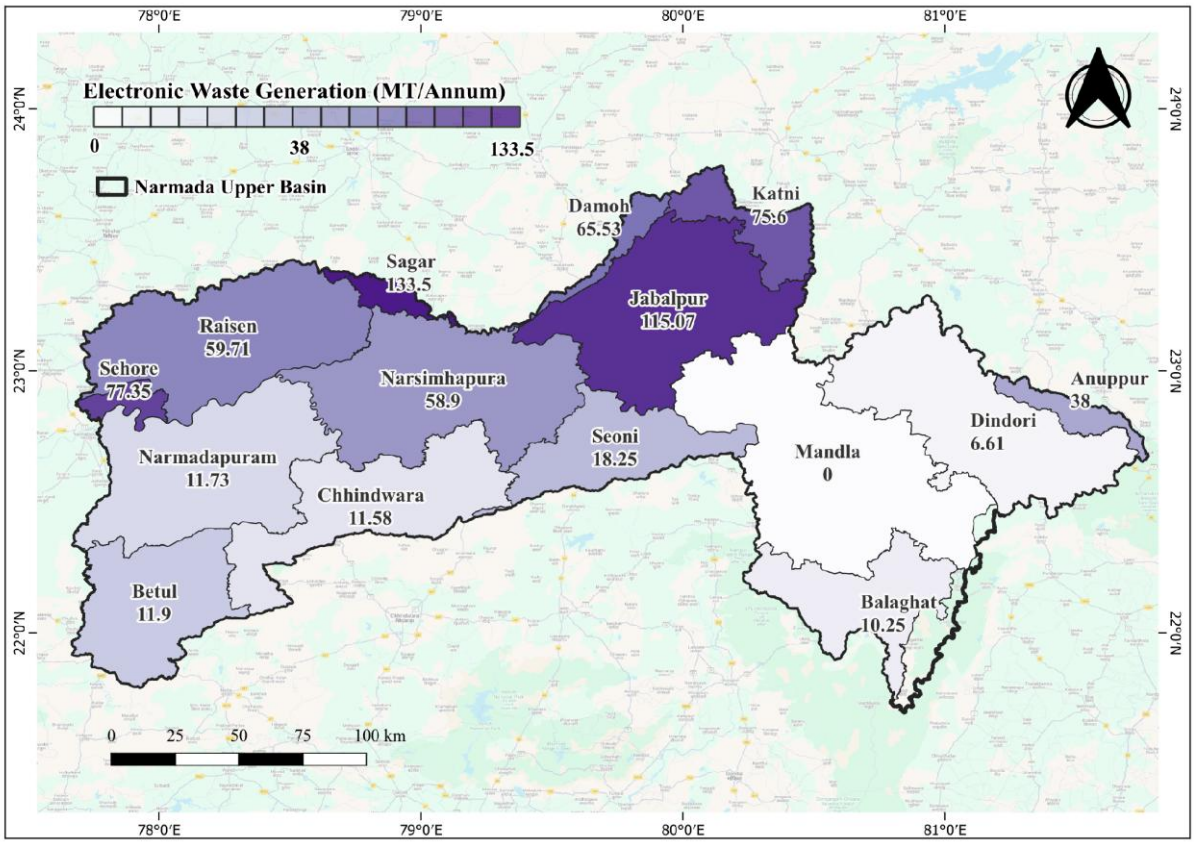


Figure 27 District-wise Electronic Waste Generation(MT/Annum) in the Narmada Upper Basin, highlighting hotspots and data gaps

(Source: Madhya Pradesh Pollution Control Board (MPPCB). (2025). District Environmental Plans (DEPs))

The Map shown in Fig. 27 offers a detailed view of electronic waste (e-waste) generation across the districts of the Narmada Upper Basin, Madhya Pradesh, India, quantified in metric tons per annum (MT/annum) and visualized by district hotspot intensity. The highest concentrations of e-waste appear in Sagar (133.5 MT/annum), Jabalpur (115.07 MT/annum), Katni (75.6 MT/annum), Sehore (77.35 MT/annum), and Damoh (65.53 MT/annum), highlighted in the darkest colors. These districts reflect urbanization patterns, higher population density, and greater penetration of electronics, underscoring the urgent need for dedicated e-waste collection facilities, formal recycling systems, and public awareness campaigns about safe disposal practices. Raisen (59.71 MT/annum), Narsimhapura (58.9 MT/annum), Chhindwara (11.58 MT/annum), Betul (11.9 MT/annum), and Seoni (18.25 MT/annum) fall into the

moderate generation category. While their outputs are not as high as principal hotspots, progressive localization of e-waste recycling infrastructure, repair initiatives, and effective involvement of informal waste pickers can prevent environmental contamination and health hazards from improper e-waste disposal. Several districts, including Balaghat (10.25 MT/annum), Narmadapuram (11.73 MT/annum), Anuppur (38 MT/annum), and Dindori (6.61 MT/annum), contribute much less e-waste, shown in lighter colors. Mandla reports 0 MT/annum, possibly due to low urbanization and minimal electronic usage. These districts can scale best practices as digital inclusion rises, focusing early on collection, refurbishment, and community sensitization measures.

Sagar, Jabalpur, Katni, Sehore, and Damoh function as e-waste generation hotspots and merit the highest priority for infrastructure provisioning and digital waste management policy. Moderate generators, including Raisen, Narsimhapura, Chhindwara, Betul, and Seoni, require scalable, community-led e-waste handling solutions. Low-output districts benefit from early engagement and inclusivity in e-waste planning, preparing them for future growth as electronic access expands. The map represents spatial intelligence for prioritizing investment and interventions around e-waste sustainability in the Narmada Upper Basin.

10.2 Middle Basin

The Map shown in Fig. 28 displays district-wise annual electronic waste (e-waste) generation across the Narmada Middle Basin, identifying major urban and semi-urban hotspots. Indore dominates e-waste generation with 51,115.53 MT/year, far surpassing all other districts, and is followed by Dewas (89.92 MT), Sehore (77.35 MT), Jhabua (66.62 MT), and Raisen (59.71 MT). These areas are heavily urbanized, with high rates of digital device consumption and disposal. Such massive figures pose urgent environmental management challenges, highlighting the need for formal collection centers, specialized processing units, and strict regulation to prevent informal or unsafe recycling. Barwani (77.2 MT), Alirajpur (41.7 MT), Burhanpur (43.4 MT), and Nandurbar (41.7 MT) represent districts in transition, where increasing adoption of electronics is rapidly enlarging e-waste streams. Public awareness campaigns and the establishment of take-back programs could arrest the growth of unsafe handling practices and boost the recovery of valuable materials. Districts such as Betul (11.9 MT), Narmadapuram (11.73 MT), Khandwa (18.73 MT), Khargone (21.6 MT), Dhar (28.56 MT), and Harda (33 MT) have comparatively lower e-waste generation rates

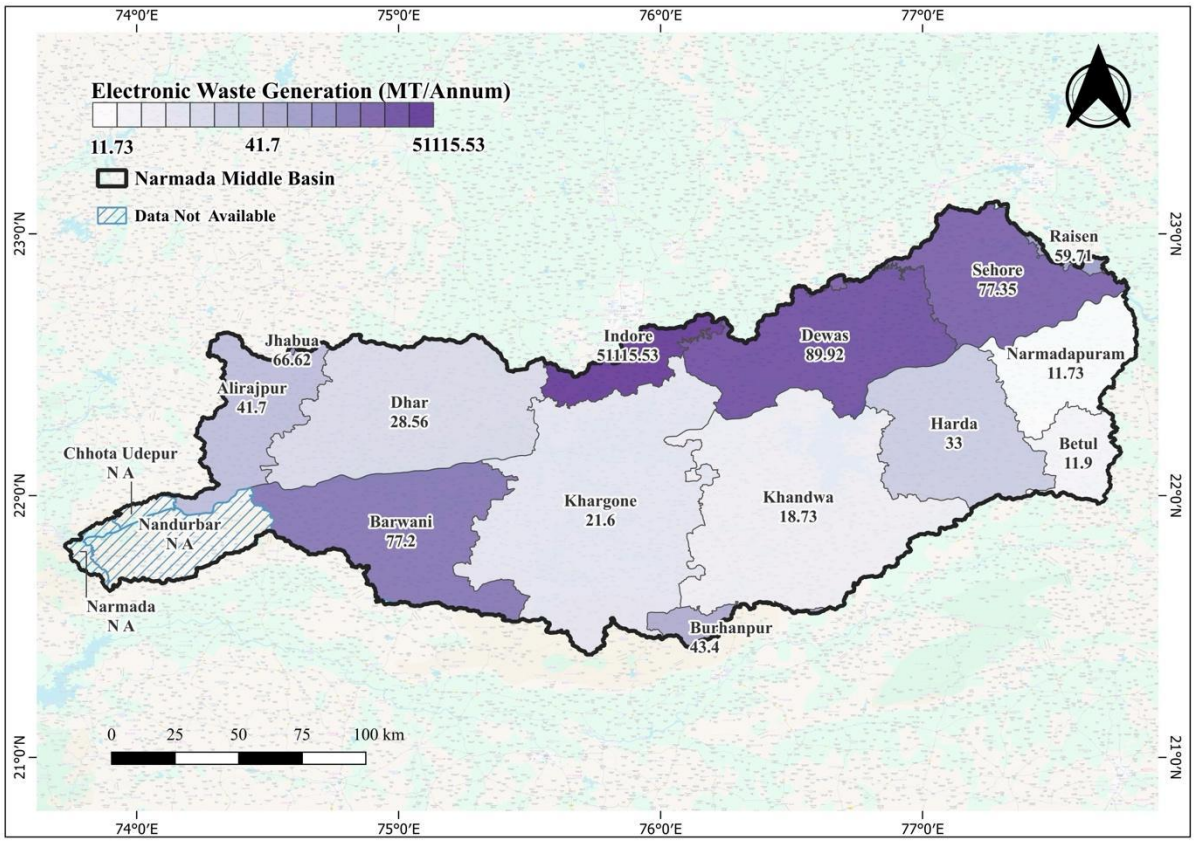


Figure 28 District-wise Electronic Waste Generation(MT/Annum) in the Narmada Middle Basin, highlighting hotspots and data gaps.

(Source: Madhya Pradesh Pollution Control Board (MPPCB). (2025). District Environmental Plans (DEPs))

Indore is by far the largest e-waste hotspot, Dewas, Sehore, Jhabua, Raisen, and Barwani are key secondary hotspots where investments in management infrastructure are crucial. Ramp-up in emerging and low-load districts by education and system-building is needed to address the issue before it scales up. Regional policies should prioritize collection, recycling, and pollution control, ensuring responsible e-waste management for human and environmental health.

11 Summary and Conclusions

The pollution load mapping of the Narmada River Basin highlights significant pressures from domestic sewage, industrial effluents, solid waste, hazardous waste, biomedical waste, plastic waste, and e-waste across both the Upper and Middle Basins. While urban centers such as Jabalpur, Sehore, and Narmadapuram exhibit high Biochemical Oxygen Demand (BOD) and fecal coliform counts, reflecting untreated sewage discharge. Jabalpur leads in domestic sewage generation, while Narsimhapur and Betul are major industrial effluent contributors, yet many districts lack adequate sewage treatment plants, resulting in direct wastewater inflows into the river. Solid waste hotspots include Jabalpur, Raisen, and Khargone, whereas hazardous waste is concentrated in Sehore, Sagar, Dhar, and Indore. Biomedical waste generation peaks in Jabalpur, Katni, and Khargone, with plastic waste highest in Mandla, Khargone, and Dewas, while Indore dominates e-waste generation. The findings emphasize the urgent need for strengthening wastewater treatment infrastructure, enforcing pollution control in industrial zones, and promoting sustainable waste management practices to safeguard the ecological and public health integrity of the Narmada River Basin.

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